Numerical Solution of Differential and Differential-Algebraic Equations (NUMDIFF-16), September 2021

A multirate variational approach to simulation and optimal control for flexible spacecraft

Yana Lishkova , DPhil Candidate*

Prof. Sina Ober-Blöbaum §

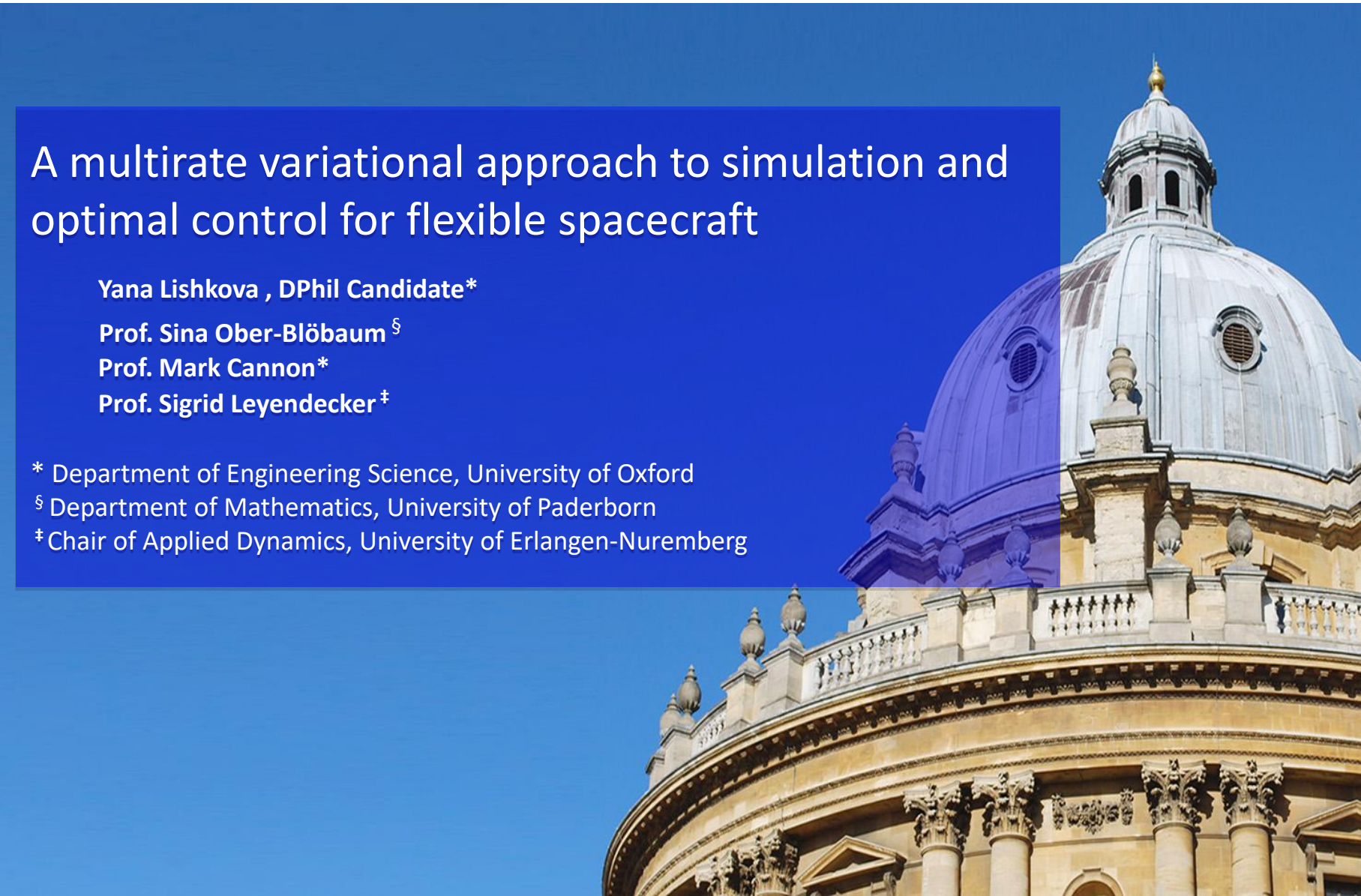
Prof. Mark Cannon*

Prof. Sigrid Leyendecker ‡

* Department of Engineering Science, University of Oxford

§ Department of Mathematics, University of Paderborn

‡ Chair of Applied Dynamics, University of Erlangen-Nuremberg



Outline

- Short overview
- Multirate DMOC
- Case study : flexible spacecraft control
- Computational results and discussion

[1] Y. Lishkova, S. Ober-Blöbaum, M. Cannon, and S. Leyendecker, "A multirate variational approach to simulation and optimal control for flexible spacecraft*", in AAS Advances in the Astronautical Sciences, Volume 175, Astrodynamics 2020*

Multirate DMOC

- **Multirate Discrete Mechanics and Optimal Control (Multirate DMOC):**

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[2] J. E. Marsden and M. West, “Discrete mechanics and variational integrators,” Acta Numerica, Vol. 10, No. 1, 2001, pp. 357–514.

[3] S. Leyendecker and S. Ober-Blöbaum, “A Variational Approach to Multirate Integration for Constrained Systems,” *Multibody Dynamics: Computational Methods and Applications* (J.-C. Samin and P. Fisette, eds.), pp. 97–121, Dordrecht: Springer Netherlands, 2013

[4] T. Gail, S. Ober-Blöbaum, and S. Leyendecker, “Variational multirate integration in discrete mechanics and optimal control,” *Proceedings of* ECCOMAS, 2017, pp. 1–4.

[5] O. Junge, J. E. Marsden, and S. Ober-Blöbaum, “Discrete mechanics and optimal control,” *IFAC Proceedings Volumes*, Vol. 38, No. 1, 2005, pp. 538–543.

Multirate DMOC

- **Multirate Discrete Mechanics and Optimal Control (Multirate DMOC):**

  High fidelity simulation at a reduced computational cost

Advantages

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  High fidelity simulation at a reduced computational cost

  - ✓ **Structure-preserving -> *conservation of energy and/or momenta of the system***

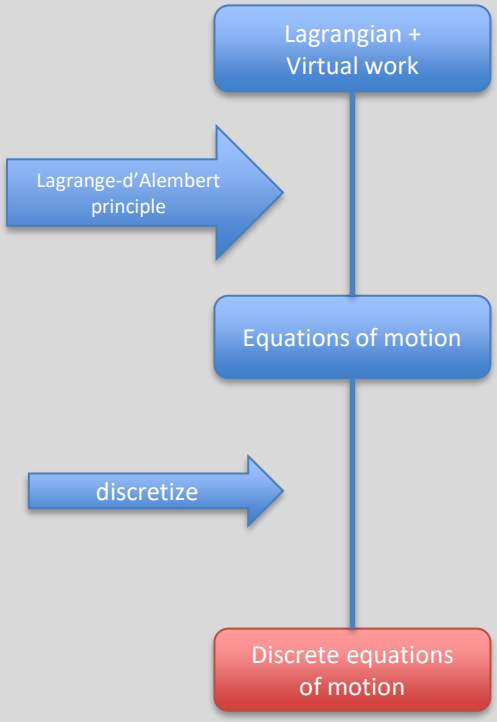
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Standard direct methods

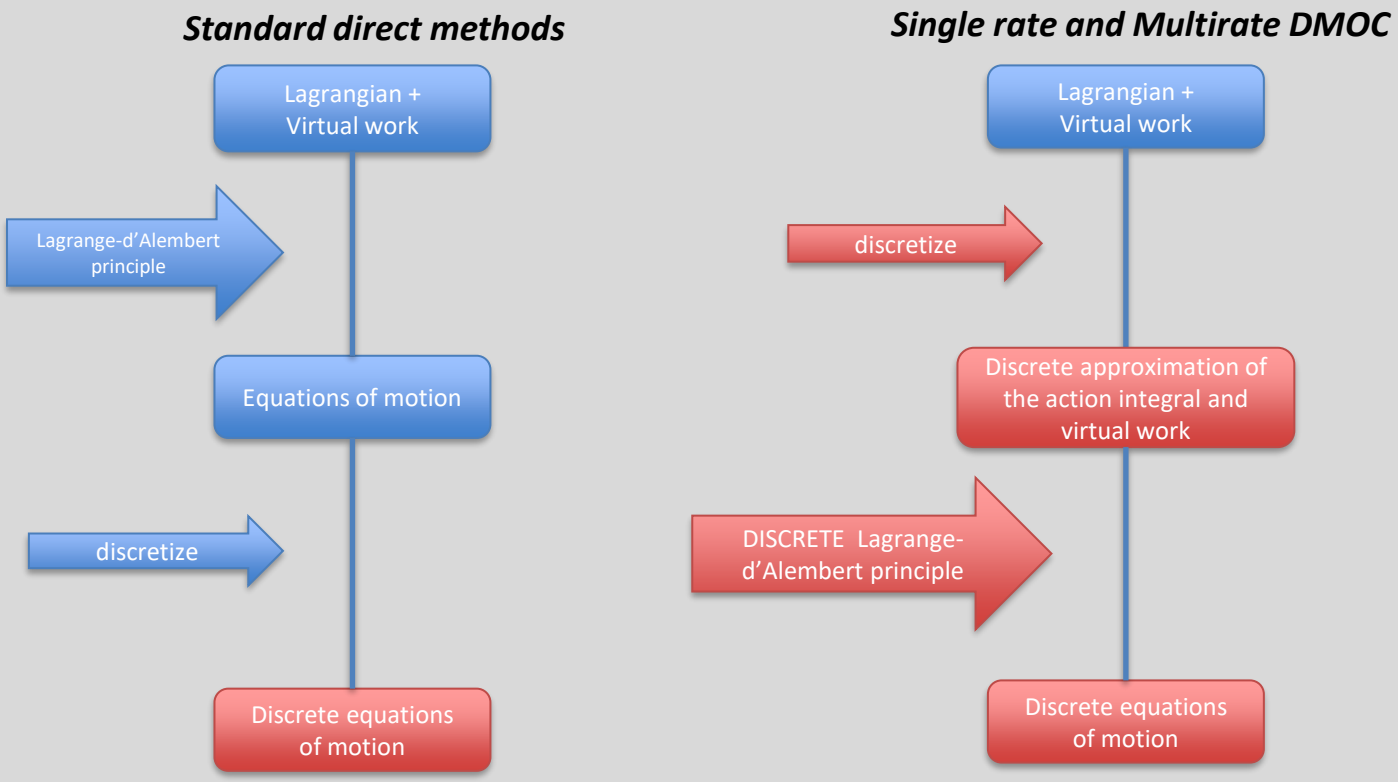


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Comparison

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action integral

$$\int_0^T L(q(t), \dot{q}(t))\,\mathrm{d}t$$

virtual work

$$\int_0^T f(q(t), \dot{q}(t)) \cdot \delta q(t)\,\mathrm{d}t$$

Lagrange- d'Alembert principle

$$\delta \int_0^1 L(q(t), \dot{q}(t))\,dt + \int_0^1 f(t) \cdot \delta q(t)\,dt = 0$$

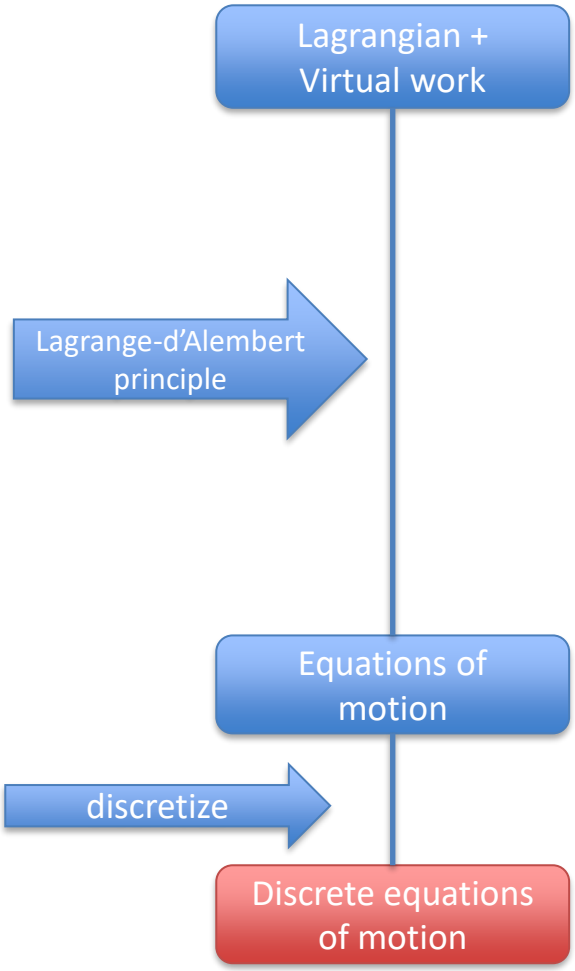
for all variations δq with $\delta q(0) = \delta q(1) = 0$

the *forced Euler-Lagrange equations*

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = f(q, \dot{q})$$

discretize the forced Euler-Lagrange equations

Standard direct methods



Comparison

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Single rate DMOC

Lagrangian + Virtual work

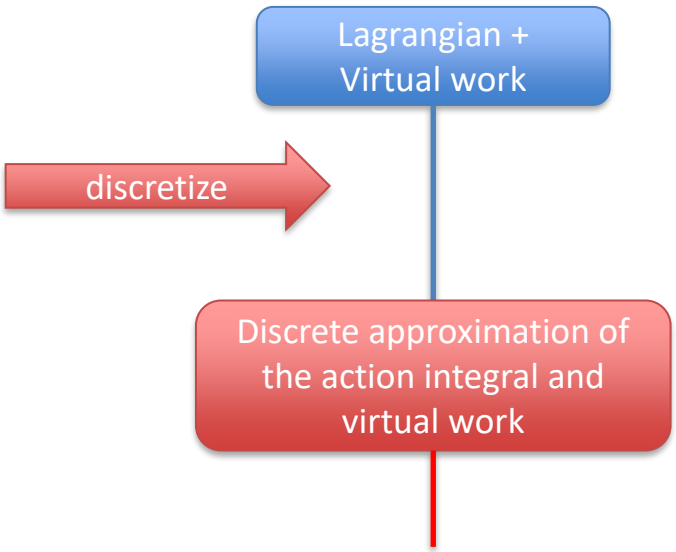
$$\int_0^T L(q(t), \dot{q}(t))\,\mathrm{d}t \qquad \textit{action integral}$$

$$\int_0^T f(q(t), \dot{q}(t)) \cdot \delta q(t)\,\mathrm{d}t \qquad \textit{virtual work}$$

Comparison

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Single rate DMOC



$$\underbrace{\int_0^T L(q(t),\dot{q}(t))\,\mathrm{d}t}_{action\ integral} \qquad \underbrace{\int_0^T f(q(t),\dot{q}(t))\cdot\delta q(t)\,\mathrm{d}t}_{virtual\ work}$$

the *discrete Lagrangian*

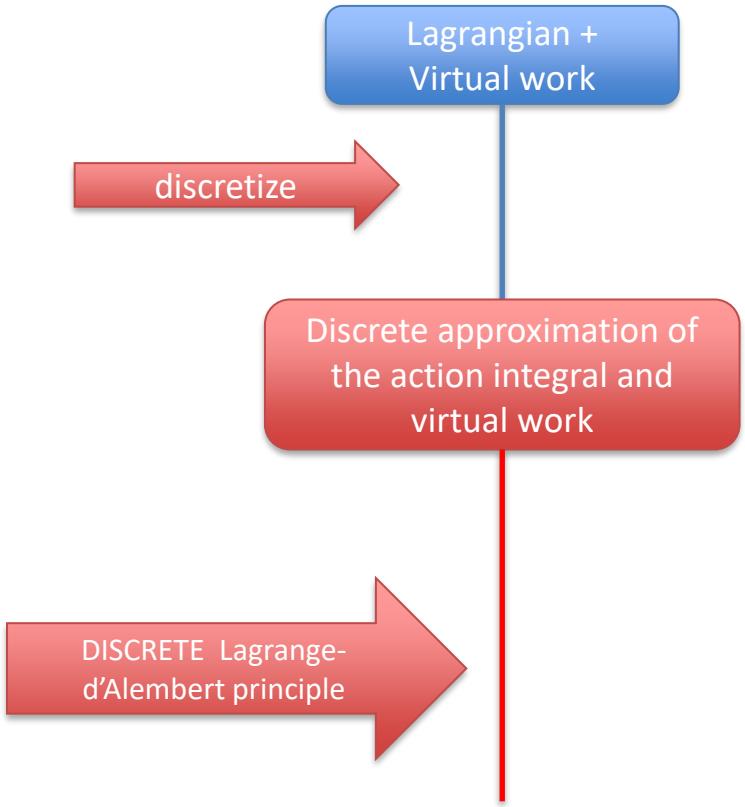
$$L_d(q_k,q_{k+1}) \approx \int_{kh}^{(k+1)h} L(q(t),\dot{q}(t))$$

virtual work approximation
by the left and right discrete forces

$$f_k^- \cdot \delta q_k + f_k^+ \cdot \delta q_{k+1} \approx \int_{kh}^{(k+1)h} f(t)\cdot\delta q(t)\ dt$$

Comparison

Single rate DMOC



$$\underbrace{\int_0^T L(q(t),\dot{q}(t))\,\mathrm{d}t}_{\textit{action integral}} \qquad \underbrace{\int_0^T f(q(t),\dot{q}(t))\cdot\delta q(t)\,\mathrm{d}t}_{\textit{virtual work}}$$

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discrete Lagrange- d'Alembert principle

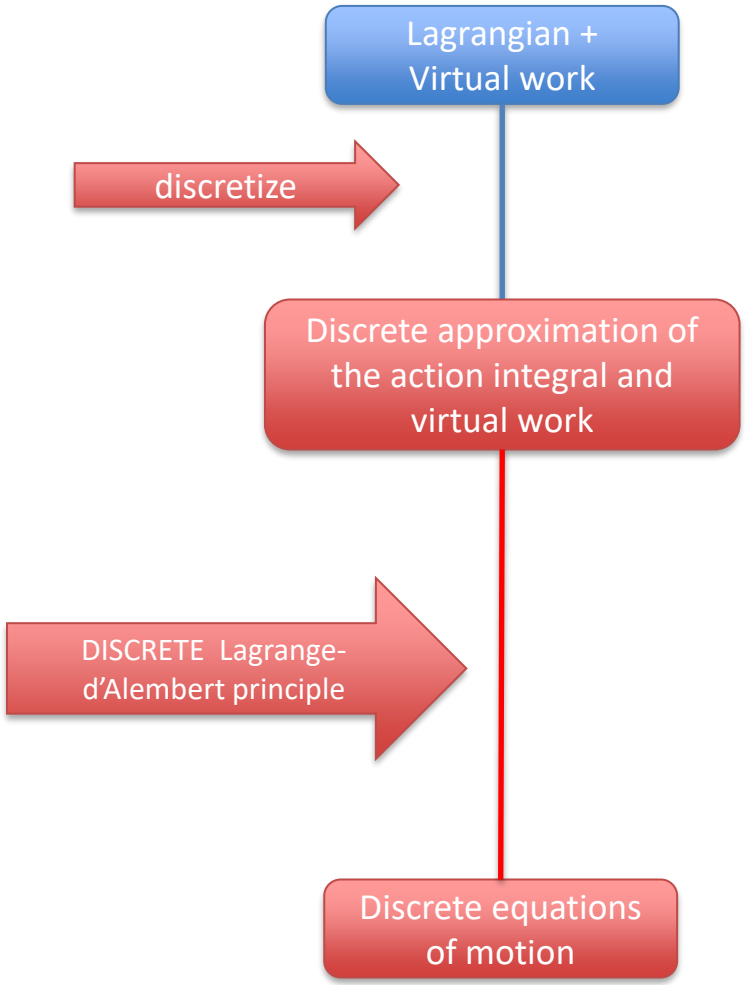
$$\delta \sum_{k=0}^{N-1} L_d(q_k,q_{k+1}) + \sum_{k=0}^{N-1} f_k^- \cdot \delta q_k + f_k^+ \cdot \delta q_{k+1} = 0$$

for all variations $\{\delta q_k\}_{k=0}^N$ with $\delta q_0 = \delta q_N = 0$

Comparison

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Single rate DMOC



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for all variations $\{\delta q_k\}_{k=0}^N$ with $\delta q_0 = \delta q_N = 0$

discrete forced Euler-Lagrange equations

$$D_2L_d(q_{k-1},q_k) + D_1L_d(q_k,q_{k+1}) + f_{k-1}^+ + f_k^- = 0$$

Comparison for simple OCP example

Standard direct methods

Discretize the following problem

$$\min_{q(\cdot),\dot{q}(\cdot),u(\cdot),(T)} J(q,\dot{q},u) = \int_0^T C(q(t),\dot{q}(t),u(t))\,\mathrm{d}t$$

subject to

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = f(q,\dot{q},u)$$

$$q(0) = q^0,\ \dot{q}(0) = \dot{q}^0,$$

Single rate DMOC

Directly obtains a structure-preserving discrete OCP of the form

$$\min_{q_d,u_d,(h)} J_d(q_d,u_d) = \sum_{k=0}^{N-1} C_d(q_k,q_{k+1},u_k)$$

subject to

$$D_2L_d(q_{k-1},q_k) + D_1L_d(q_k,q_{k+1}) + f_{k-1}^+ + f_k^- = 0$$

for $k = 1,\ldots,N-1$

$$q_0 = q^0,\ D_2L(q^0,\dot{q}^0) + D_1L_d(q_0,q_1) + f_0^- = 0$$

Advantages

- **Multirate Discrete Mechanics and Optimal Control (Multirate DMOC):**

  High fidelity simulation at a reduced computational cost

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  - ✓ **Multirate discretization -> *computational efficiency*** [7]

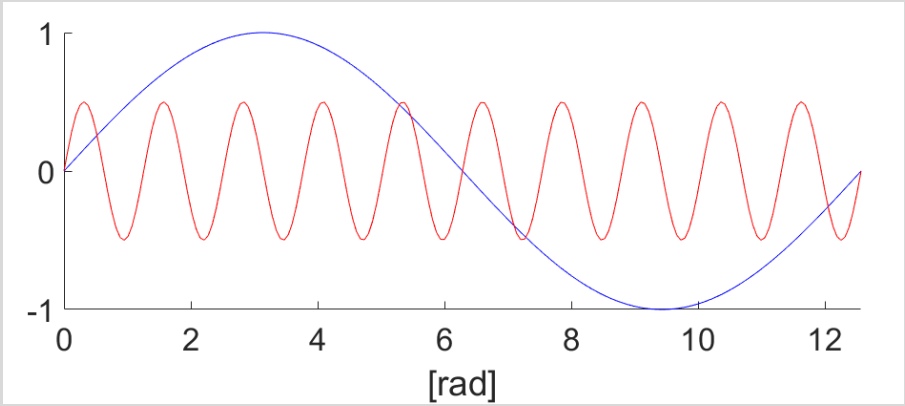
[7] T. Gail, S. Leyendecker, and S. Ober-Blöbaum, "Computing time investigations of variational multirate integrators," *ECCOMAS Multibody Dynamics*, 2013.

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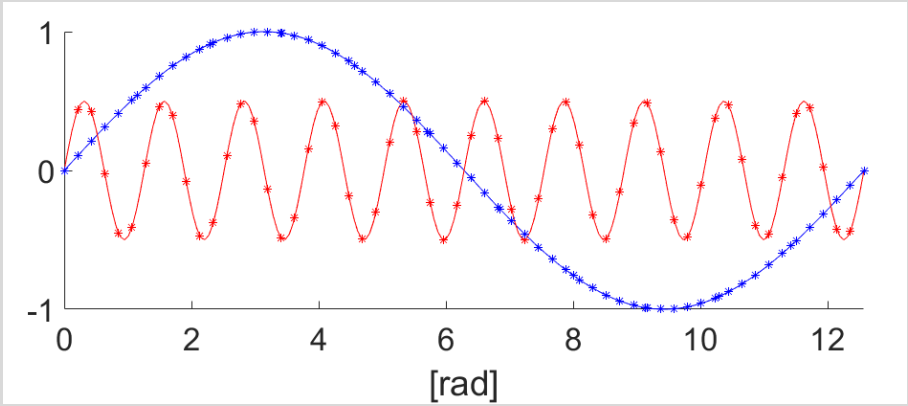


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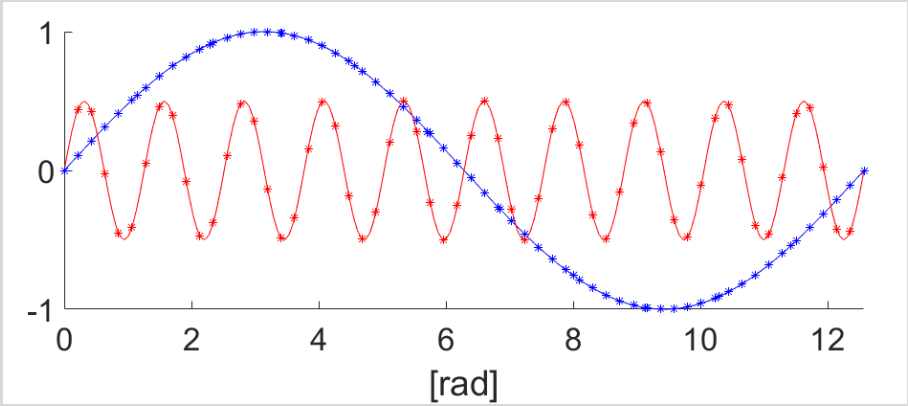
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system dynamics

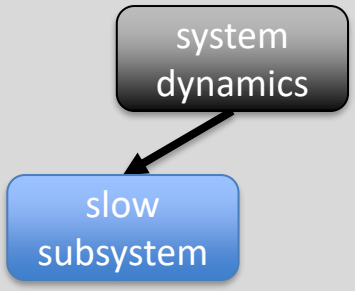


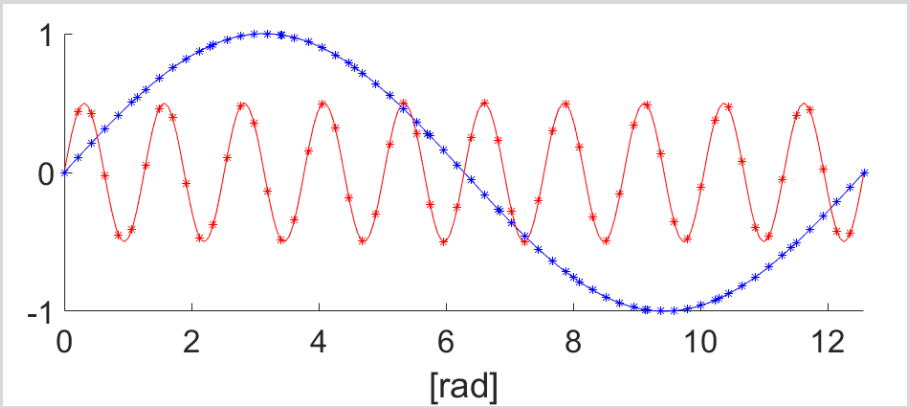
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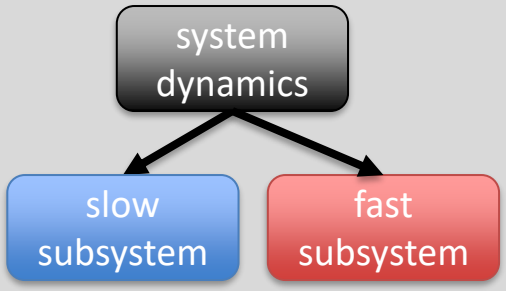


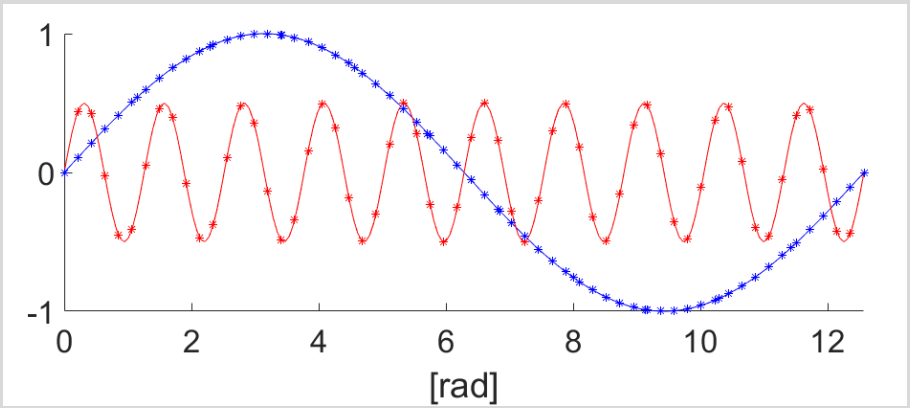
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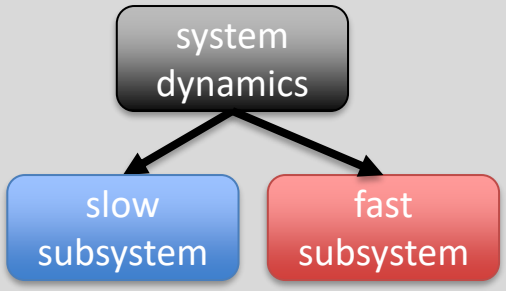


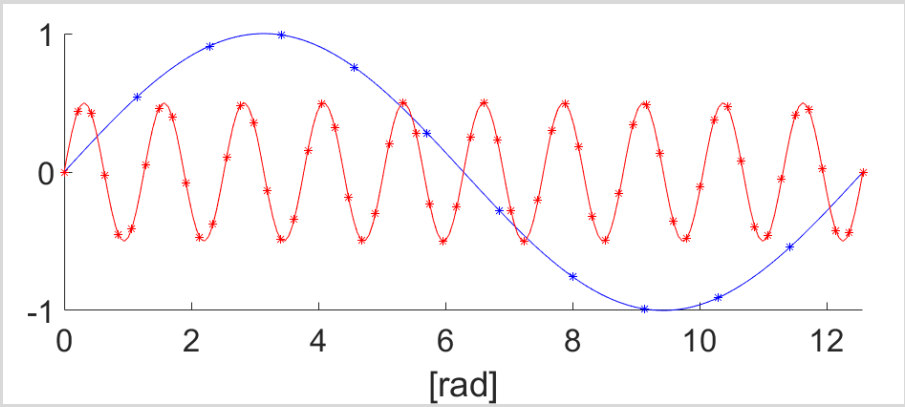
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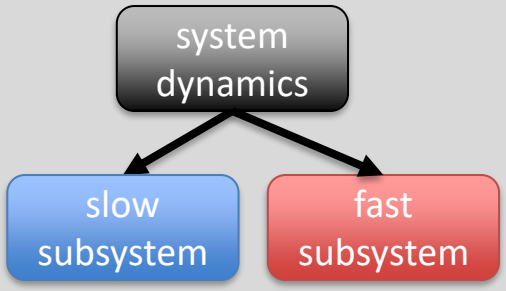
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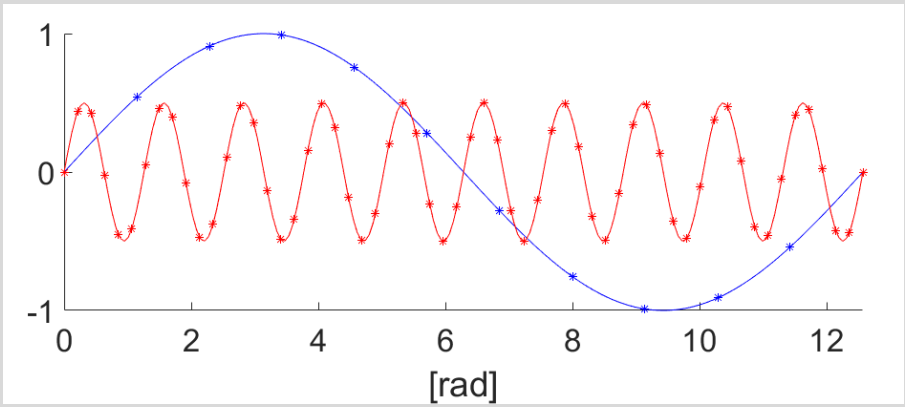
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- **Reduces the number of optimization variables**
- **Reduces the number of equality constraints**

Multirate formulation

If this system exhibits dynamics on different scales we assume that the vector of generalised coordinates can be separated or transformed into slow and fast variables as:

$$\underline{q}(t) = \begin{bmatrix} \underline{q}^s(t) \\ \underline{q}^f(t) \end{bmatrix} \qquad \text{or} \qquad \underline{q}(t) = g(\underline{q}^s(t), \underline{q}^f(t))$$

Where: $\underline{q} \in \mathbb{R}^n$ $\underline{q}^s \in \mathbb{R}^{n_s}$ $\underline{q}^f \in \mathbb{R}^{n_f}$ $n = n_s + n_f$

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The Lagrangian can be rewritten in the new coordinates and if possible for the given example, the stiff potential could also be separated:

$$L(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) = T(\dot{\underline{q}}^s, \dot{\underline{q}}^f) - V(\underline{q}^s, \underline{q}^f) - W(\underline{q}^f)$$

Where $T(\dot{\underline{q}}^s, \dot{\underline{q}}^f)$ represents the kinetic energy of the system and the potential energy of the system takes the form $U(\underline{q}^s, \underline{q}^f) = V(\underline{q}^s, \underline{q}^f) + W(\underline{q}^f)$

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Based on this coordinate change the external forces can be resolved as:

$$\underline{f}\,(\underline{q}^s, \underline{q}^f) \cdot (\delta\underline{q}^s, \delta\underline{q}^f) = \underline{f}^s \cdot \delta\underline{q}^s + \underline{f}^f \cdot \delta\underline{q}^f$$

Comparison for simple *multirate* OCP example

Standard direct methods

Discretize the following *multirate* problem

$$\min_{\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}^s, \underline{u}^f} J(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}^s, \underline{u}^f) = \int_0^T C(\underline{q}^s(t), \underline{q}^f(t), \dot{\underline{q}}^s(t), \dot{\underline{q}}^f(t), \underline{u}^s, \underline{u}^f) dt$$

subject to

$$D_1\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) - \frac{d}{dt}\Big(D_3\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f)\Big) + f^s(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}) = 0$$
$$D_2\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) - \frac{d}{dt}\Big(D_4\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f)\Big) + f^f(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}) = 0$$

$$(\underline{q}^s(0), \dot{\underline{q}}^s(0)) = (\underline{q}^{s,0}, \dot{\underline{q}}^{s,0}), \quad (\underline{q}^f(0), \dot{\underline{q}}^f(0)) = (\underline{q}^{f,0}, \dot{\underline{q}}^{f,0})$$
$$(\underline{q}^s(T), \dot{\underline{q}}^s(T)) = (\underline{q}^{s,T}, \dot{\underline{q}}^{s,T}), \quad (\underline{q}^f(T), \dot{\underline{q}}^f(T)) = (\underline{q}^{f,T}, \dot{\underline{q}}^{f,T})$$

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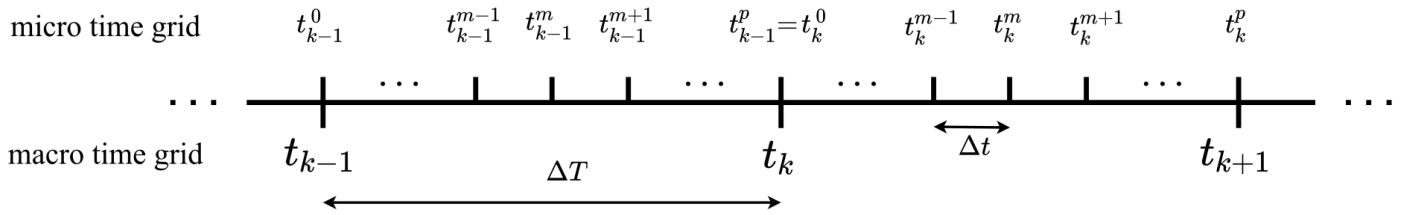
$$D_1\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) - \frac{d}{dt}\Big(D_3\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f)\Big) + f^s(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}) = 0$$

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$$p = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



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Standard direct methods

Discretize the following *multirate* problem

$$\min_{\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}^s, \underline{u}^f} J(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}^s, \underline{u}^f) = \int_0^T C(\underline{q}^s(t), \underline{q}^f(t), \dot{\underline{q}}^s(t), \dot{\underline{q}}^f(t), \underline{u}^s, \underline{u}^f)dt$$

subject to

$$D_1\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) - \frac{d}{dt}\Big(D_3\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f)\Big) + f^s(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}) = 0$$
$$D_2\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f) - \frac{d}{dt}\Big(D_4\mathcal{L}(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f)\Big) + f^f(\underline{q}^s, \underline{q}^f, \dot{\underline{q}}^s, \dot{\underline{q}}^f, \underline{u}) = 0$$

$$(\underline{q}^s(0), \dot{\underline{q}}^s(0)) = (\underline{q}^{s,0}, \dot{\underline{q}}^{s,0}), \quad (\underline{q}^f(0), \dot{\underline{q}}^f(0)) = (\underline{q}^{f,0}, \dot{\underline{q}}^{f,0})$$
$$(\underline{q}^s(T), \dot{\underline{q}}^s(T)) = (\underline{q}^{s,T}, \dot{\underline{q}}^{s,T}), \quad (\underline{q}^f(T), \dot{\underline{q}}^f(T)) = (\underline{q}^{f,T}, \dot{\underline{q}}^{f,T})$$

Multirate DMOC

Directly obtains a structure-preserving discrete OCP of the form

$$\min_{\underline{q}_d^s, \underline{q}_d^f, \underline{u}_d^s, \underline{u}_d^f} J_d(\underline{q}_d^s, \underline{q}_d^f, \underline{u}_d^s, \underline{u}_d^f) = \sum_{k=0}^{N-1} C_d^k(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f, \underline{u}_{k+1/2}^s, \underline{u}_k^f)$$

subject to

$$D_{\underline{q}_k^s}(L_d^k + L_d^{k-1}) + F_k^{s-} + F_{k-1}^{s,+} = 0 \qquad \text{for } k = 1, \dots, N-1$$
$$D_{\underline{q}_k^{f,0}}(L_d^k - L_d^{k-1}) + \underline{f}_k^{f,0,-} + \underline{f}_k^{f,p-1,+} = 0 \qquad \text{for } k = 1, \dots, N-1$$
$$D_{\underline{q}_k^{f,m}} L_d^k + \underline{f}_k^{f,m,-} + \underline{f}_k^{f,m-1,+} = 0 \qquad \text{for } k = 0, \dots, N-1, \; m = 1, \dots, p-1$$

$$\underline{q}_0^s = \underline{q}^s(0), \quad \underline{q}_0^{f,0} = \underline{q}^f(0), \qquad \underline{q}_N^s = \underline{q}^s(T), \quad \underline{q}_{N-1}^{f,p} = \underline{q}^f(T),$$
$$\underline{p}_0^s = \underline{p}^s(0), \quad \underline{p}_0^{f,0} = \underline{p}^f(0), \qquad \underline{p}_N^s = \underline{p}^s(T), \quad \underline{p}_{N-1}^{f,p} = \underline{p}^f(T)$$

$$\underline{p}_0^s = -D_{\underline{q}_0^s} L_d^0(\underline{q}_0^s, \underline{q}_1^s, \underline{q}_0^f) - F_0^{s,-}$$
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$$\underline{p}_N^s = D_{\underline{q}_N^s} L_d^{N-1}(\underline{q}_{N-1}^s, \underline{q}_N^s, \underline{q}_{N-1}^f) + F_{N-1}^{s,+}$$
$$\underline{p}_{N-1}^{f,p} = D_{\underline{q}_{N-1}^{f,p}} L_d^{N-1,p-1}(\underline{q}_{N-1}^s, \underline{q}_N^s, \underline{q}_{N-1}^f) + \underline{f}_{N-1}^{f,p-1,+}$$

Advantages

- **Multirate Discrete Mechanics and Optimal Control (Multirate DMOC):**

  High fidelity simulation at a reduced computational cost

  - ✓ Structure-preserving -> *conservation of energy and/or momenta of the system*
  - ✓ Multirate discretization -> *computational efficiency*

[8] M. Azadi, M. Eghtesad, S. Fazelzadeh, and E. Azadi, "Dynamics and control of a smart flexible satellite moving in an orbit," *Multibody System Dynamics*, Vol. 35, No. 1, 2015, pp. 1–23.

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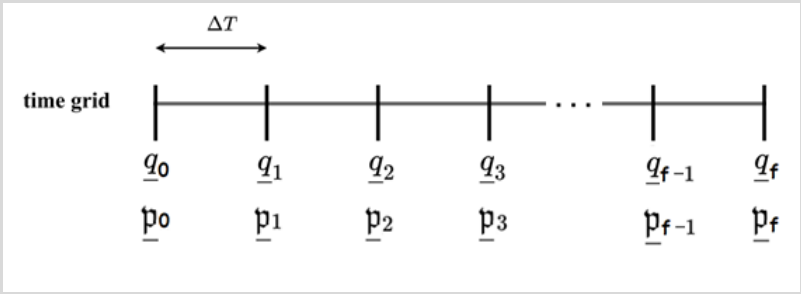
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Advantages

- ***DMOC has two equivalent formulations:***

  - Integration in the phase space

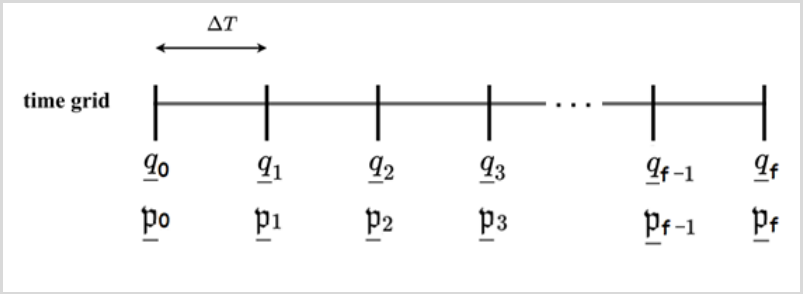


Integrates both the configuration and the momentum to obtain the system trajectory in discrete time

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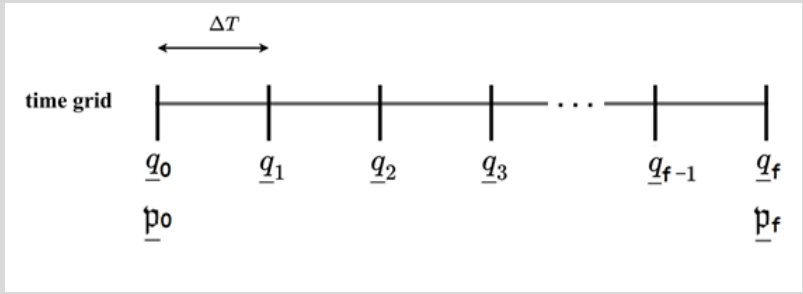
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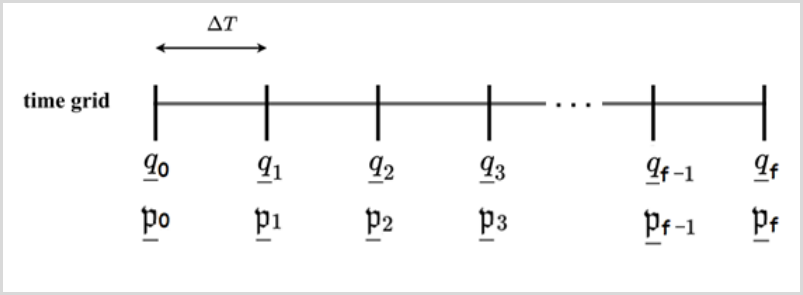


Avoids computation of the momentum on all nodes apart from the boundaries while retaining the structure-preserving properties.

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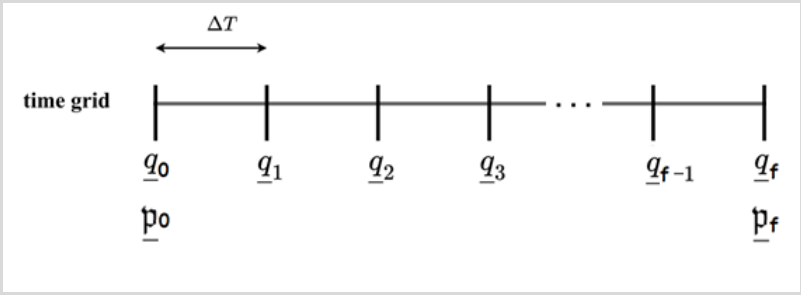
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In the discrete OCP this reduces the number of constraints by approximately a half and the number of unknowns by minimum of a third leading to further substantial computational savings.

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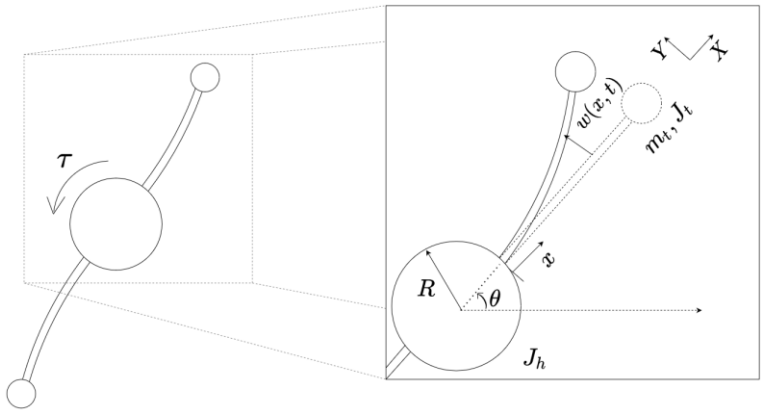
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The mathematical model

- **General linear model neglecting dissipation effects**
  - **single-axis rest-to-rest rotational maneuver**
  - **control torque applied at the hub**

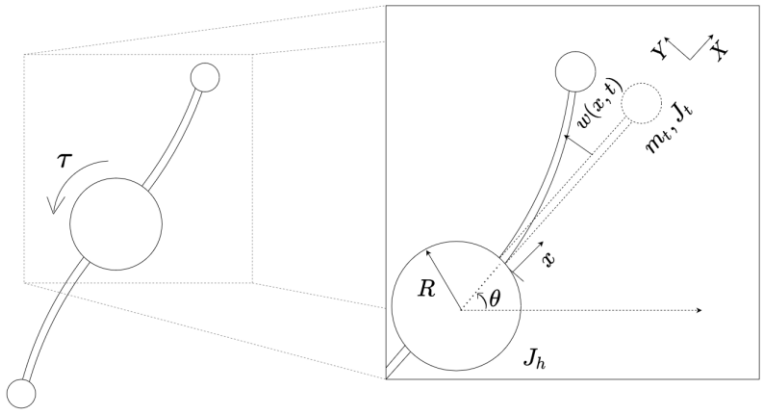


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- **Spatial discretization - The Assumed Modes Method** [9]

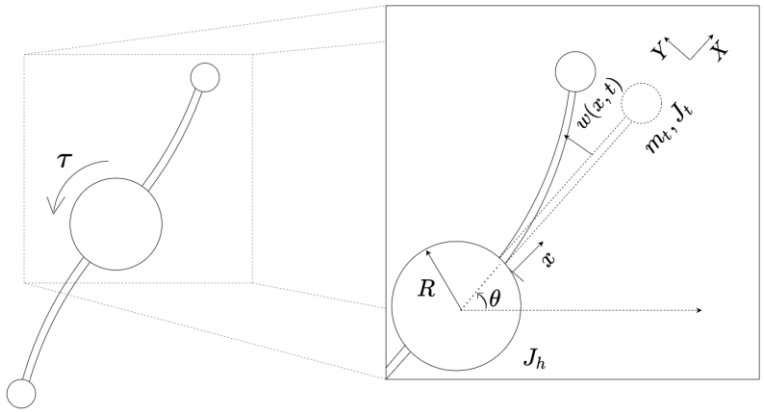
$$w(x,t) = \sum_{j=1}^{N} \phi_j(x)\, \eta_j(t), \qquad x \in [0, L]$$



ϕ_j - assumed spatial mode shapes
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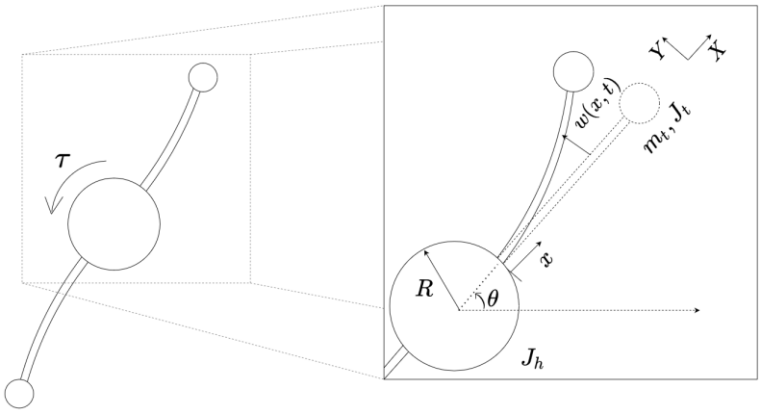
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- **System description [7]**

$$\underline{\xi} = \begin{bmatrix} \theta \\ \underline{\eta} \end{bmatrix}, \qquad \underline{\eta} = [\eta_1,\ \eta_2,\ \ldots,\ \eta_N]^T \in R^{N\times 1}$$

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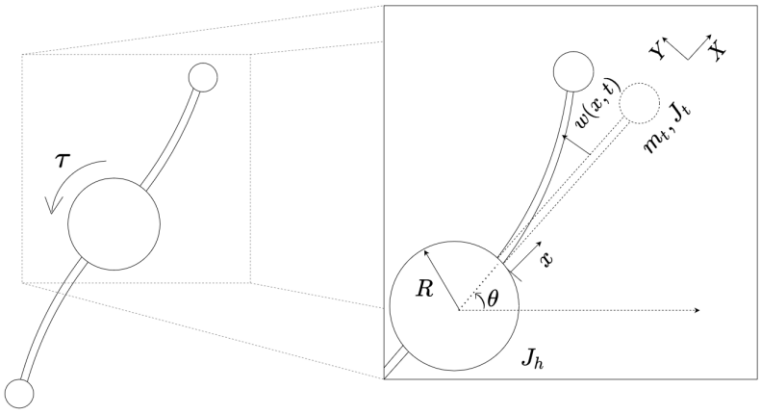
  - **Lagrange - d'Alembert principle**

$$\mathcal{L}(\underline{\xi},\, \dot{\underline{\xi}}) = \frac{1}{2}\, \dot{\underline{\xi}}^T\, \mathbf{M}\, \dot{\underline{\xi}} - \frac{1}{2}\, \underline{\xi}^T\, \mathbf{K}\, \underline{\xi}, \qquad \delta W = \underline{\mathrm{f}} \cdot \underline{\xi} = \tau \cdot \delta\theta - \mathbf{C}\, \dot{\underline{\xi}} \cdot \delta\underline{\xi}$$

$$\delta \int_{t_0}^{t_f} \mathcal{L}(\underline{\xi},\, \dot{\underline{\xi}})\, dt + \int_{t_0}^{t_f} (\,\underline{\mathrm{f}} \cdot \delta\underline{\xi}\,)\, dt = 0 \text{ for all variations } \delta\underline{\xi} \text{ with } \delta\underline{\xi}(t_0) = \delta\underline{\xi}(t_f) = 0$$

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$$\mathbf{M}\,\ddot{\underline{\xi}} + \mathbf{C}\,\dot{\underline{\xi}} + \mathbf{K}\,\underline{\xi} = \mathbf{D}\,\tau\,, \quad \mathbf{M} = \begin{bmatrix} M_{\theta\theta} & M_{\theta\eta}^T \\ M_{\theta\eta} & M_{\eta\eta} \end{bmatrix},\ \mathbf{K} = \begin{bmatrix} 0 & 0 \\ 0 & K_{\eta\eta} \end{bmatrix},\ \mathbf{C} = \begin{bmatrix} C_{\theta\theta} & 0 \\ 0 & C_{\eta\eta} \end{bmatrix},\ \mathbf{D} = [1, 0, ..., 0]^T$$

Transformation to modal coordinates

$$\underline{\xi} = \begin{bmatrix} \theta \\ \underline{\eta} \end{bmatrix}$$

$$\mathcal{L}(\underline{\xi}, \dot{\underline{\xi}}) = \frac{1}{2} \dot{\underline{\xi}}^T \mathbf{M} \dot{\underline{\xi}} - \frac{1}{2} \underline{\xi}^T \mathbf{K} \underline{\xi}$$

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$$\mathbf{E}^T\mathbf{M}\mathbf{E} = \mathbf{I}, \quad \mathbf{E}^T\mathbf{K}\mathbf{E} = \mathbf{\Lambda},$$

$$w_1 \leq w_2 \leq \ldots \leq w_{N+1},$$

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Table 1: Structural parameters used for the simulations

Hub radius	R	1.0	ft
Hub rotary inertia	J_h	8.0	slug-ft^2
Tip mass	m_t	0.156941	slug
Tip mass rotary inertia	J_t	0.0018	slug-ft^2
Beam length	L	4.0	ft
Beam height	h	6.0	in.
Beam thickness	t	0.125	in.
Beam linear density	ρA	0.0271875	slug/ft
Beam elastic modulus	E	0.1584×10^{10}	lb/ft^2

Table 2: Natural frequencies

w_1	0	rad/s
w_2	6.454	rad/s
w_3	52.41	rad/s
w_4	1.607×10^2	rad/s
w_5	3.381×10^2	rad/s
w_6	5.78×10^2	rad/s

$$N = 5$$

Multirate formulation

$$\underline{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix} \begin{matrix} \} \ \underline{q}_s \longrightarrow \begin{matrix} w_1 = 0 \text{ rad/s} \\ w_2 = 6.454 \text{ rad/s} \\ w_3 = 52.41 \text{ rad/s} \end{matrix} \\ \} \ \underline{q}_f \longrightarrow \begin{matrix} w_4 = 1.607 \times 10^2 \text{ rad/s} \\ w_5 = 3.381 \times 10^2 \text{ rad/s} \\ w_6 = 5.78 \times 10^2 \text{ rad/s} \end{matrix} \end{matrix}$$

$$\mathcal{L} = \frac{1}{2}\Big((\dot{\underline{q}}^s)^T \dot{\underline{q}}^s - (\underline{q}^s)^T \mathbf{\Lambda_s}\, \underline{q}^s \Big) + \frac{1}{2}\Big((\dot{\underline{q}}^f)^T \dot{\underline{q}}^f - (\underline{q}^f)^T \mathbf{\Lambda_f}\, \underline{q}^f \Big)$$

$$\text{where } \mathbf{\Lambda_s} = \text{diag}(\omega_1^2, \ \dots \ , \omega_3^2), \ \mathbf{\Lambda_f} = \text{diag}(\omega_4^2, \ \dots \ , \omega_6^2)$$

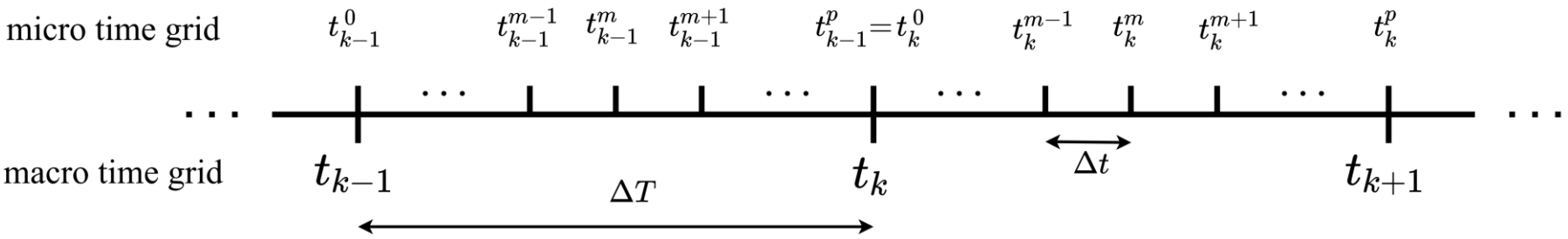
Multirate formulation

$$\underline{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix} \begin{matrix} \} \, \underline{q}_s \longrightarrow \begin{matrix} w_1 = 0 \text{ rad/s} \\ w_2 = 6.454 \text{ rad/s} \\ w_3 = 52.41 \text{ rad/s} \end{matrix} \\ \} \, \underline{q}_f \longrightarrow \begin{matrix} w_4 = 1.607 \times 10^2 \text{ rad/s} \\ w_5 = 3.381 \times 10^2 \text{ rad/s} \\ w_6 = 5.78 \times 10^2 \text{ rad/s} \end{matrix} \end{matrix}$$

$$\mathcal{L} = \frac{1}{2}\Big((\dot{\underline{q}}^s)^T \dot{\underline{q}}^s - (\underline{q}^s)^T \mathbf{\Lambda_s}\, \underline{q}^s \Big) + \frac{1}{2}\Big((\dot{\underline{q}}^f)^T \dot{\underline{q}}^f - (\underline{q}^f)^T \mathbf{\Lambda_f}\, \underline{q}^f \Big)$$

where $\mathbf{\Lambda_s} = \text{diag}(\omega_1^2, \ \dots \ , \omega_3^2)$, $\mathbf{\Lambda_f} = \text{diag}(\omega_4^2, \ \dots \ , \omega_6^2)$

$$p = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$

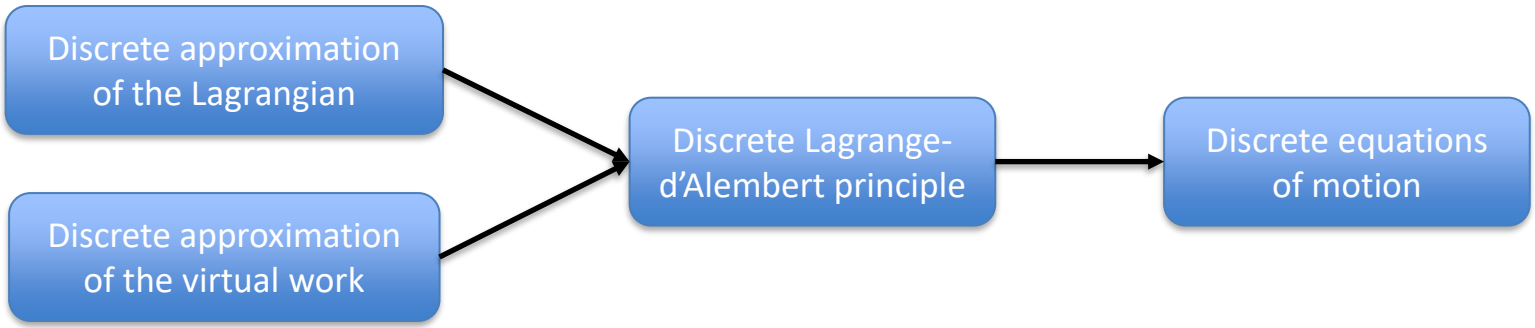


Multirate formulation

$$\underline{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{bmatrix} \begin{matrix} \} \ \underline{q}_s \longrightarrow \begin{matrix} w_1 = 0 \text{ rad/s} \\ w_2 = 6.454 \text{ rad/s} \\ w_3 = 52.41 \text{ rad/s} \end{matrix} \\ \} \ \underline{q}_f \longrightarrow \begin{matrix} w_4 = 1.607 \times 10^2 \text{ rad/s} \\ w_5 = 3.381 \times 10^2 \text{ rad/s} \\ w_6 = 5.78 \times 10^2 \text{ rad/s} \end{matrix} \end{matrix}$$

$$\mathcal{L} = \frac{1}{2}\Big((\dot{\underline{q}}^s)^T \dot{\underline{q}}^s - (\underline{q}^s)^T \mathbf{\Lambda_s}\, \underline{q}^s \Big) + \frac{1}{2}\Big((\dot{\underline{q}}^f)^T \dot{\underline{q}}^f - (\underline{q}^f)^T \mathbf{\Lambda_f}\, \underline{q}^f \Big)$$

where $\mathbf{\Lambda_s} = \text{diag}(\omega_1^2, \ \dots \ , \omega_3^2)$, $\mathbf{\Lambda_f} = \text{diag}(\omega_4^2, \ \dots \ , \omega_6^2)$



Forward simulation without control

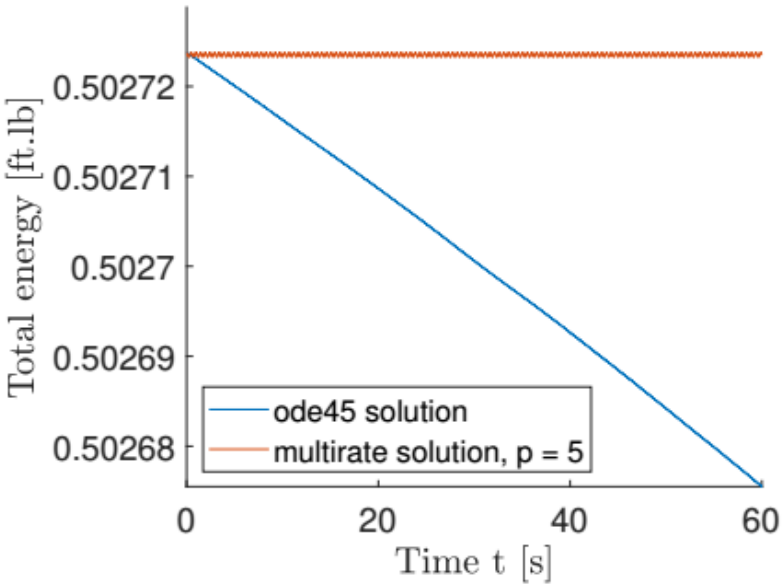


Figure 1: Numerical dissipation of total energy for simulations with $\Delta t = 10^{-4}$ and $t_f = 60$s

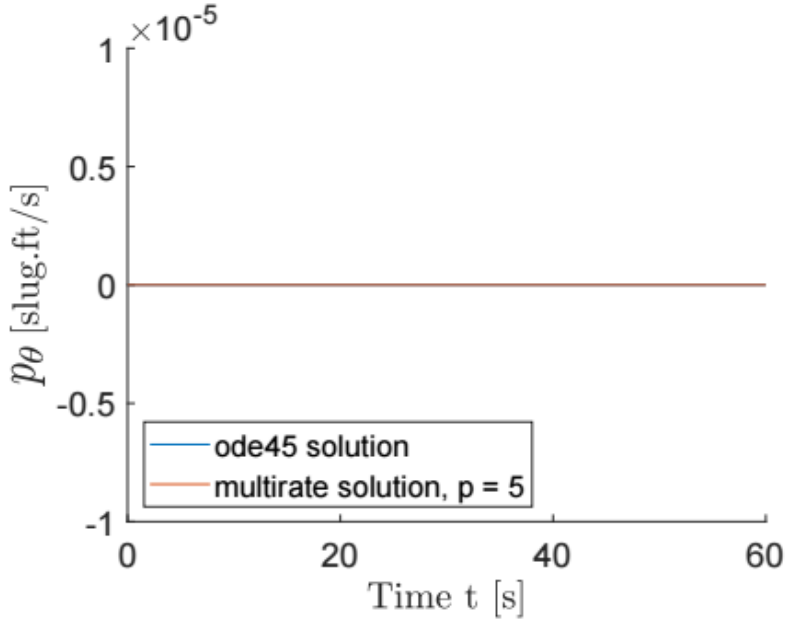


Figure 2: Momentum preservation for simulations with $\Delta t = 10^{-4}$ and $t_f = 60$s

$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = M_{\theta\theta}\,\dot{\theta} + (M_{\theta\eta})^T\,\underline{\dot{\eta}}$$

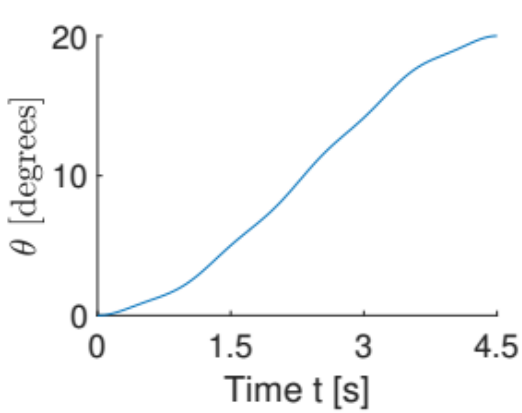
Optimal control problem formulation

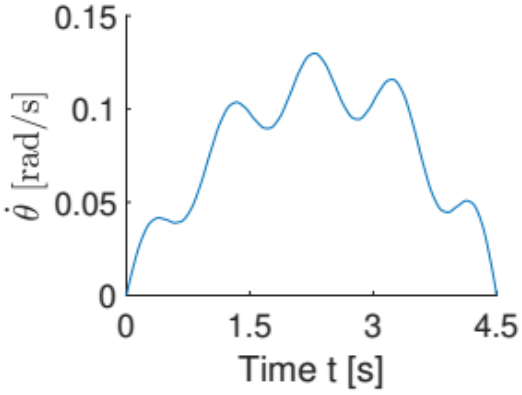
$$J(x,u) = \int_{t_0}^{t_f} C(\underline{x}(t), u(t))\, dt = \frac{1}{2}\int_{t_0}^{t_f} [\underline{x}(t)^T \mathbf{W} \underline{x}(t) + u(t)^2]\, dt$$

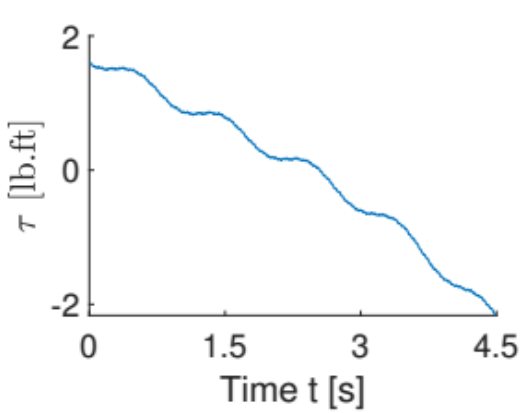
$$\text{subject to} \begin{cases} \dot{\underline{x}}(t) = \mathbf{A}\underline{x}(t) + \mathbf{B}\tau(t) \\ \underline{q}(t_0) = \mathbf{E}^{-1}\underline{\xi}_{t_0}, \quad \underline{\xi}_{t_0} = [0, ..., 0]^T \\ \underline{q}(t_f) = \mathbf{E}^{-1}\underline{\xi}_{t_f}, \quad \underline{\xi}_{t_f} = [\theta_{t_f}, 0, ..., 0]^T \\ \dot{\underline{q}}(t_0) = \dot{\underline{q}}(t_f) = [0, ..., 0]^T \end{cases}$$

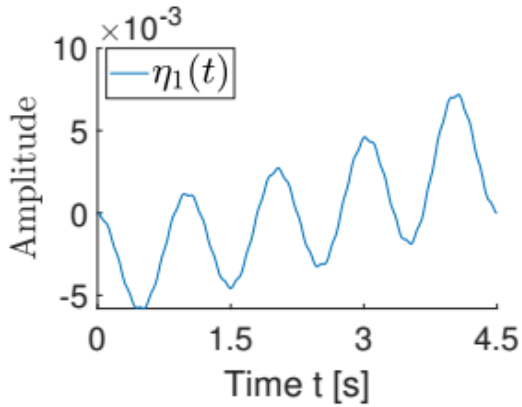
$$\text{where} \quad \underline{x}(t) = \begin{bmatrix} \underline{q}(t) \\ \dot{\underline{q}}(t) \end{bmatrix}, \quad u(t) = \tau(t), \quad \mathbf{A} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{\Lambda} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \mathbf{E}^T\mathbf{D} \end{bmatrix}, \quad \mathbf{W} = \mathbf{I}$$

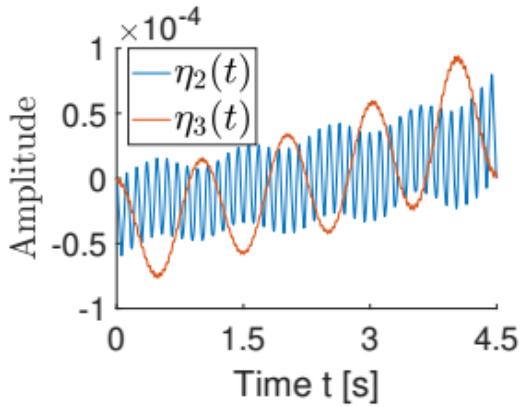
Example solution with phase space formulation











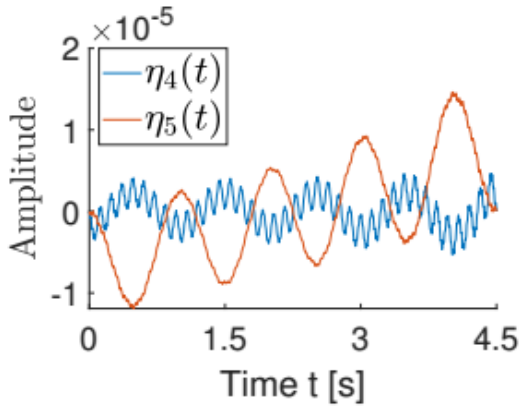
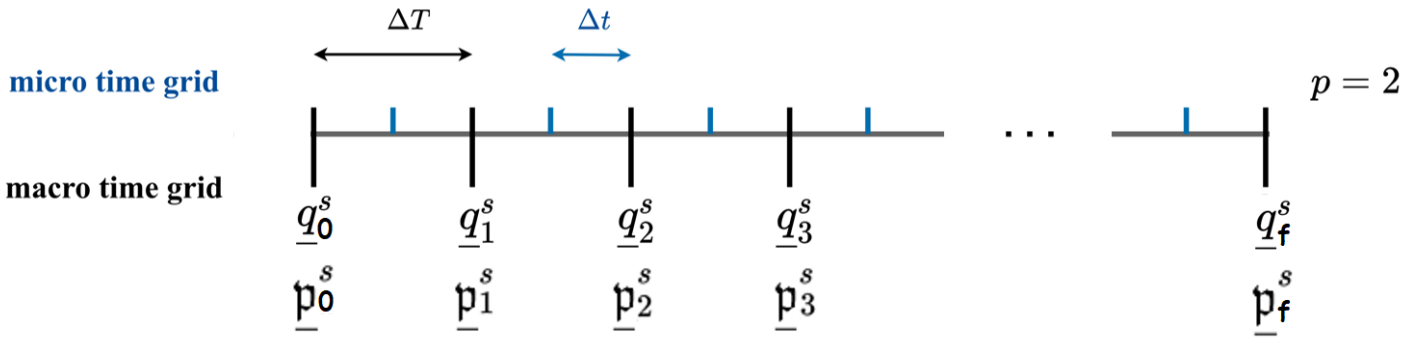


Figure 3: OCP solution with Multirate DMOC for $\theta_{t_f} = 20°$, $t_f = 4.5s$, $\Delta t = 10^{-3}$ and $p = 5$

Problem size

$$\underline{\mathfrak{p}} = \frac{\partial \mathcal{L}}{\partial \dot{\underline{q}}}$$

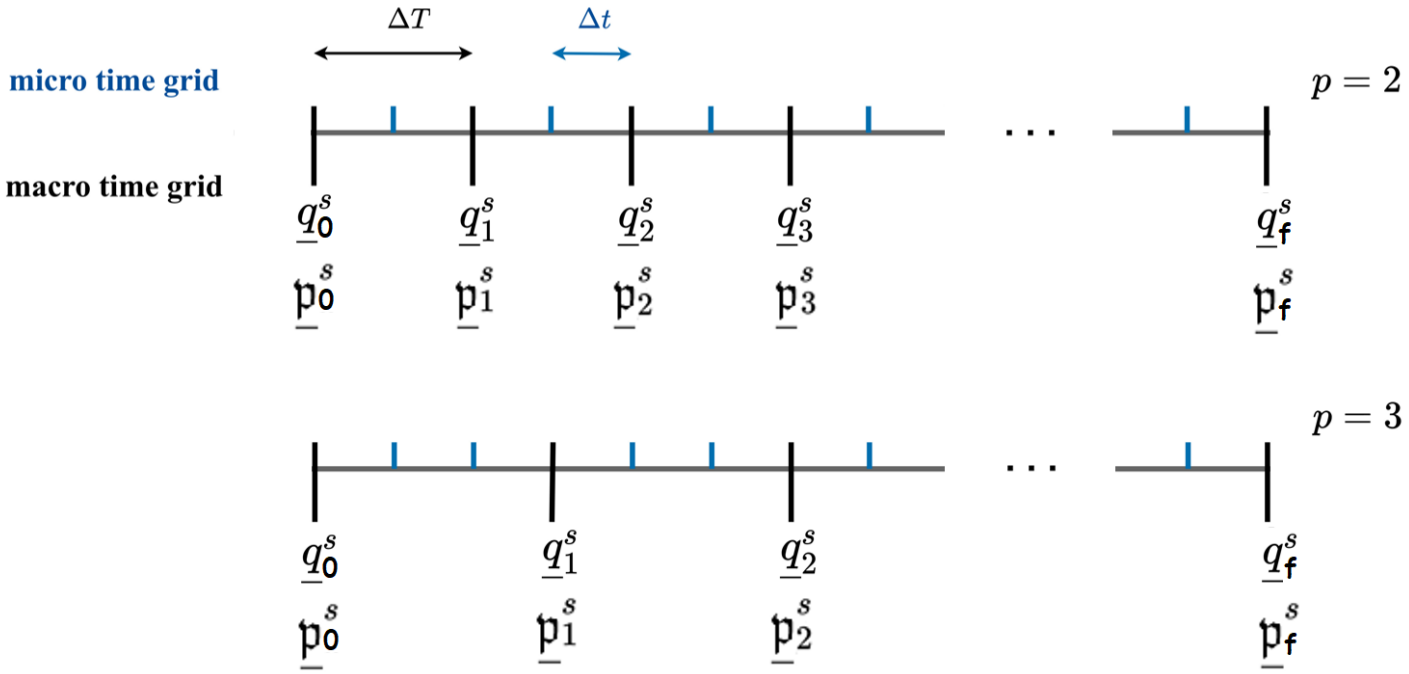
$$p = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



Problem size

$$\underline{\mathfrak{p}} = \frac{\partial \mathcal{L}}{\partial \dot{\underline{q}}}$$

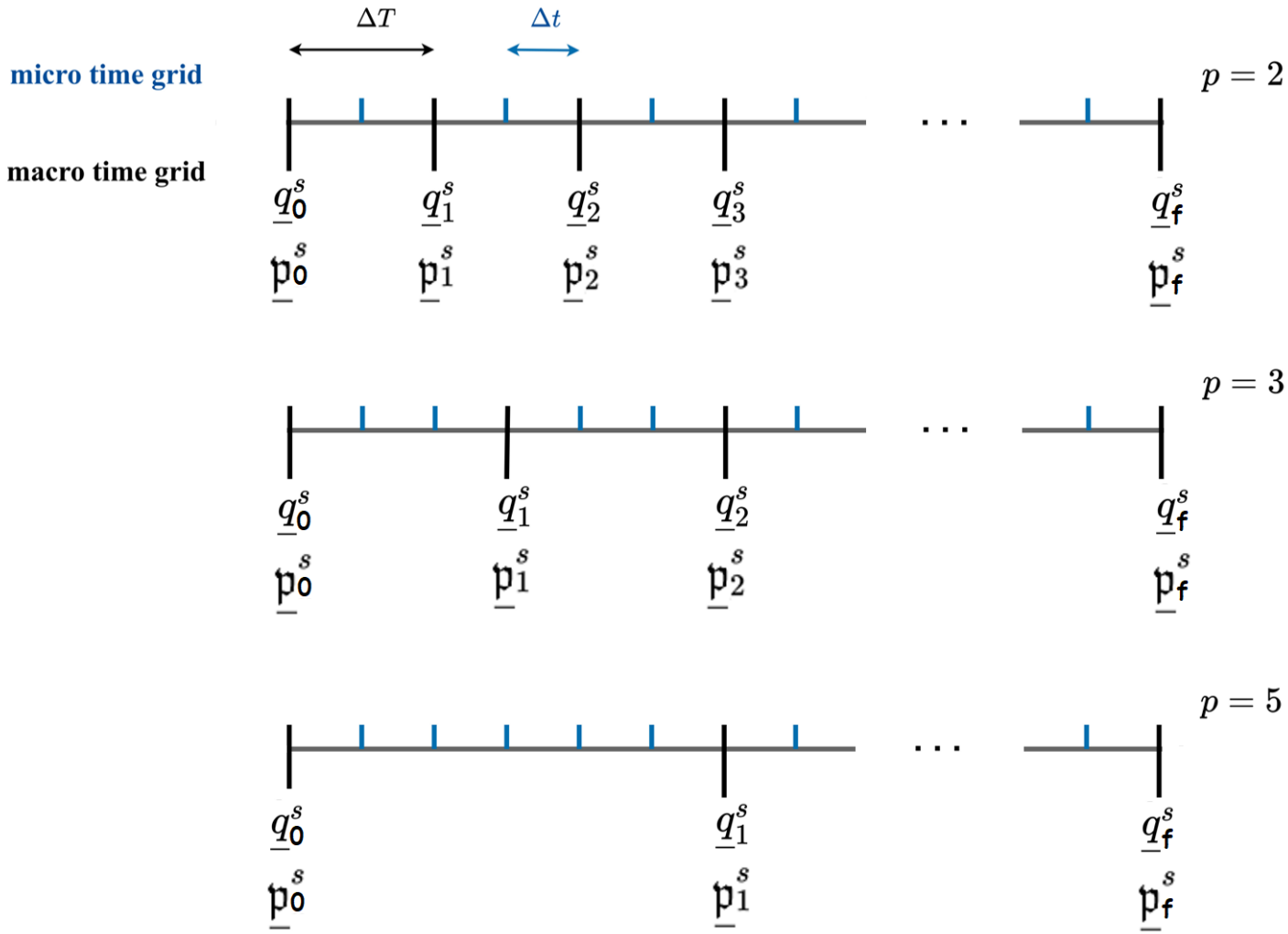
$$p = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



Problem size

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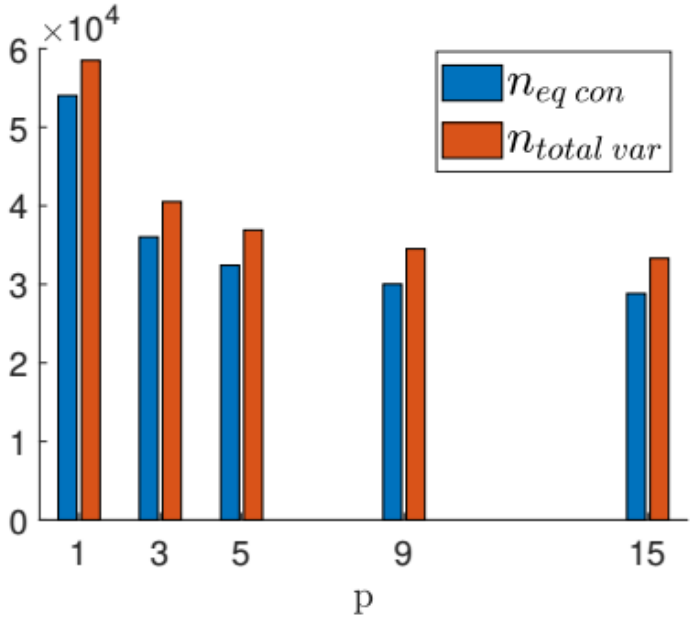
Problem size

$n_{eq\,con}$ - number of equality constraints

$n_{slow\,var}$ - number of optimization variables resulting from discretization on the macro grid

$n_{fast\,var}$ - number of optimization variables resulting from discretization on the micro grid

$n_{total\,var} = n_{slow\,var} + n_{fast\,var}$



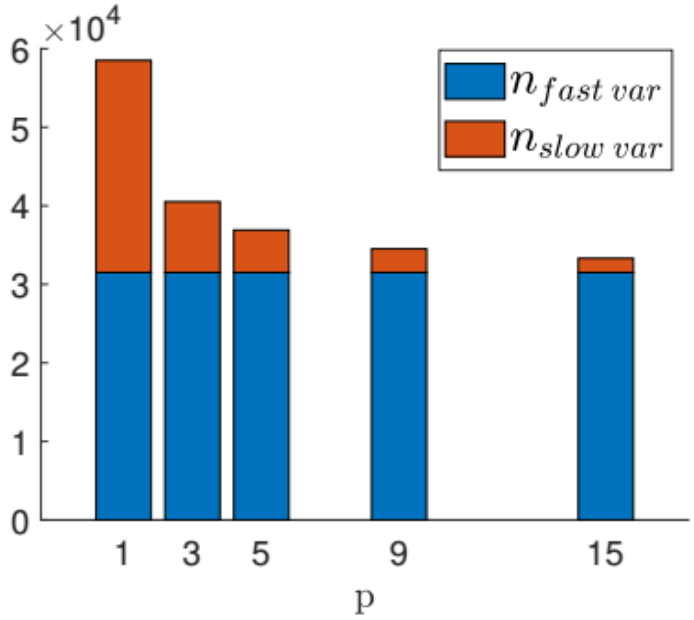


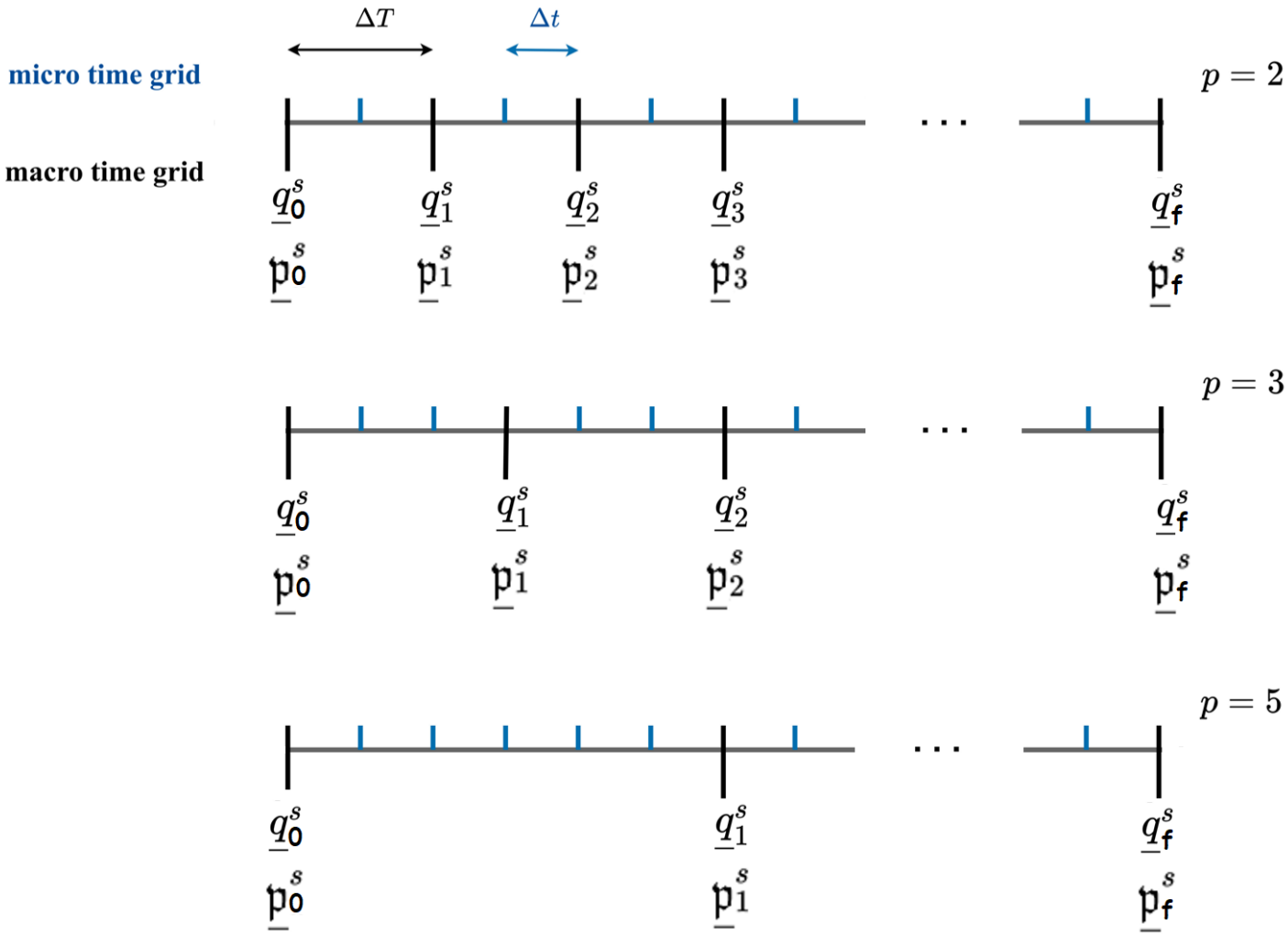
Figure 4: Size of OCP based on Multirate DMOC for a simulations with $\Delta t = 10^{-3}$ and $t_f = 4.5s$

Problem size

$$\underline{\mathfrak{p}} = \frac{\partial \mathcal{L}}{\partial \dot{\underline{q}}}$$

$$p = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$

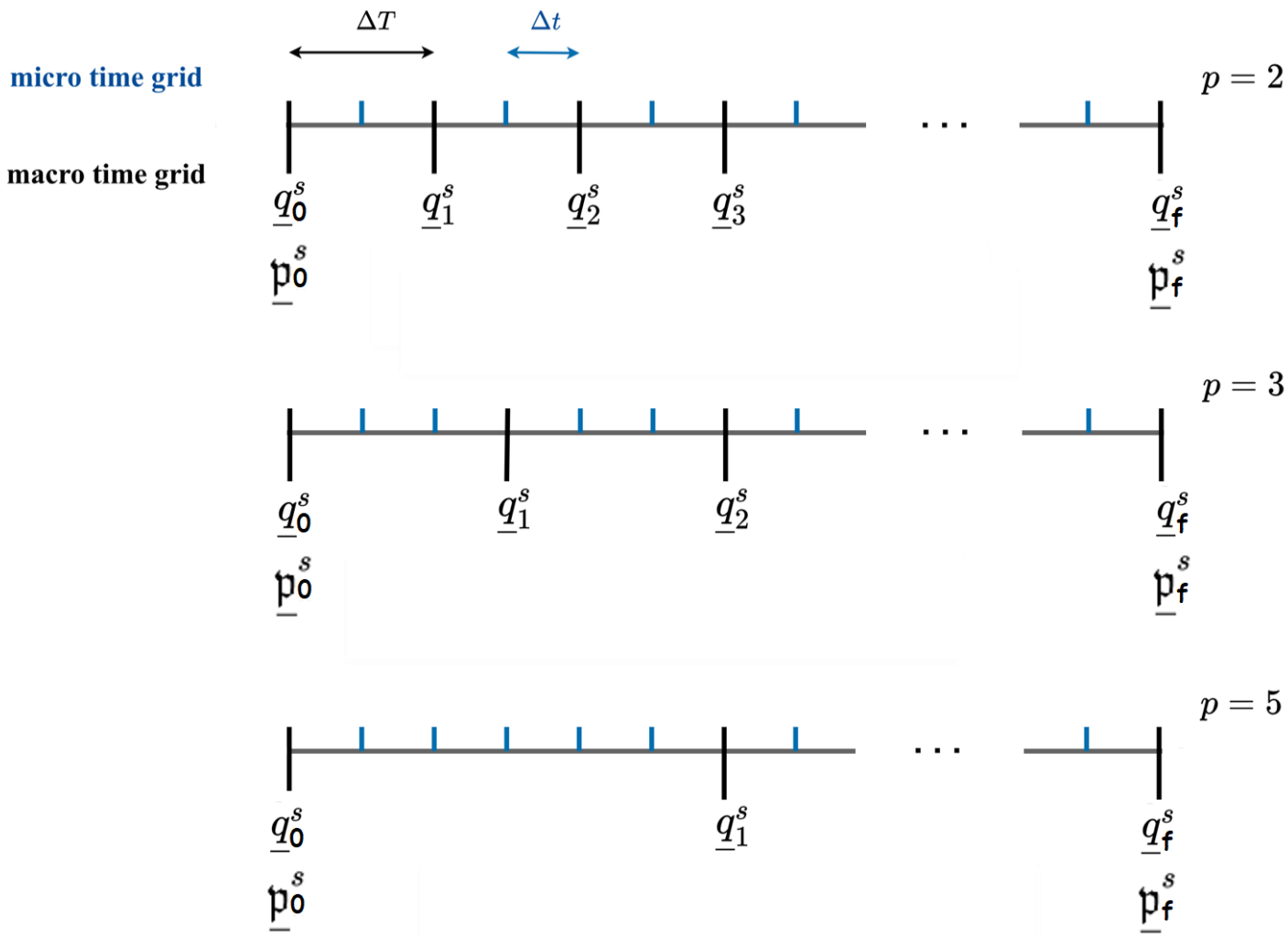
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Problem size

$$\underline{\mathfrak{p}} = \frac{\partial \mathcal{L}}{\partial \underline{\dot{q}}}$$

$$p = \frac{\text{macro time step}}{\text{micro time step}} = \frac{\Delta T}{\Delta t}$$



Problem size:

Formulation with momentum computation

Formulation without momentum computation

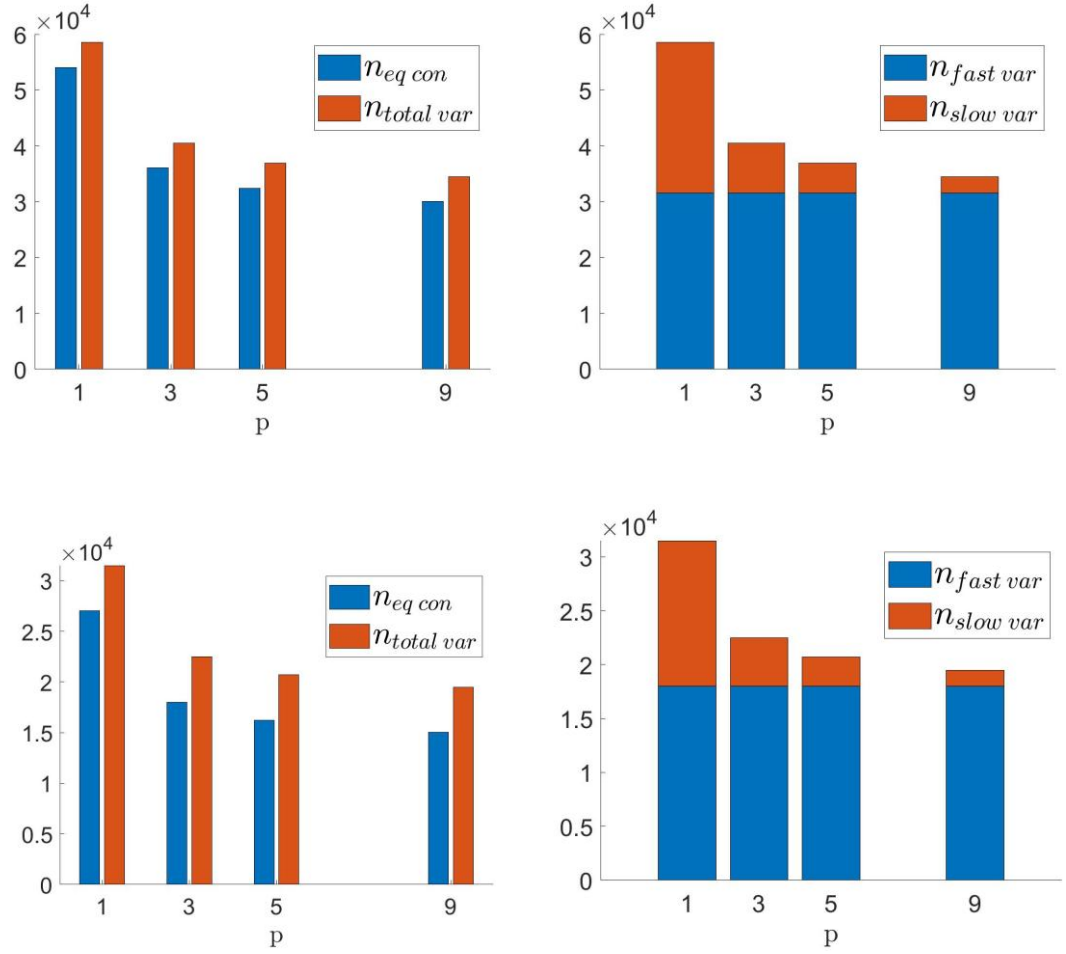
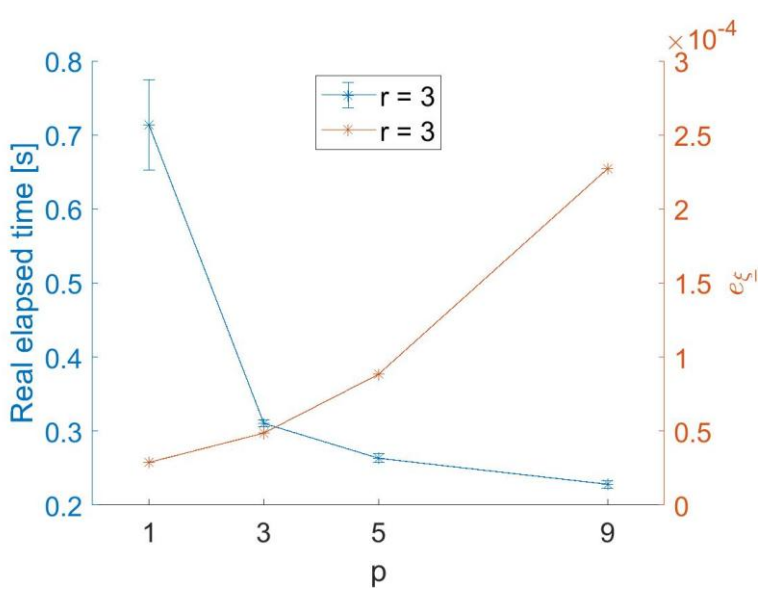


Figure 7: Size of the OCP based on the two formulations of Multirate DMOC with $\Delta t = 10^{-4}$, $t_f = 0.45s$ and no cost penalty for the momentum $\left(\mathbf{W} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & 0 \end{bmatrix}\right)$

Main result- the trade-off:

Formulation with momentum computation



Formulation without momentum computation

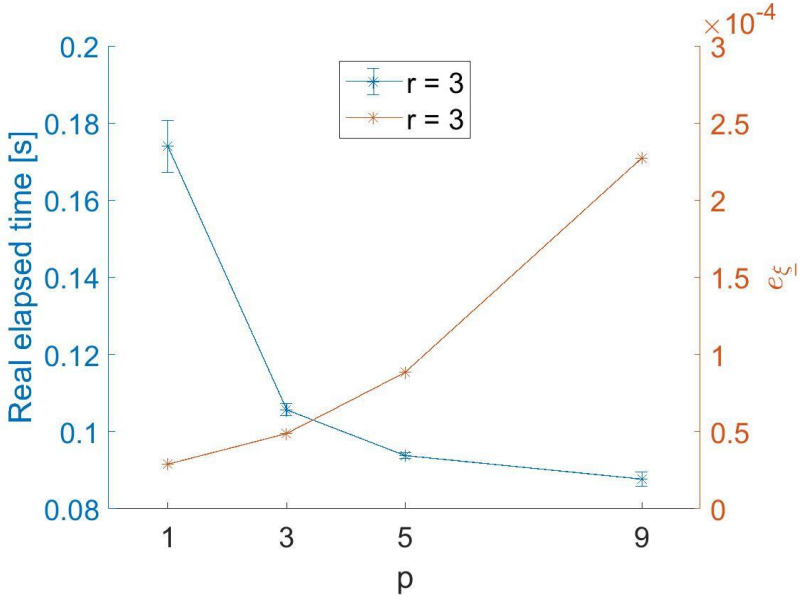


Figure 8: Mean computational time with standard deviation and relative error in ξ versus p for constant micro time step $\Delta t = 10^{-4}$, $t_f = 0.45s$ and no cost penalty for the momentum $\left(\mathbf{W} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & 0 \end{bmatrix} \right)$

Further customization

$\underline{q}^s \in R^{r\times 1}$ r - degrees of freedom of the slow subsystem

$\underline{q}^f \in R^{(N+1-r)\times 1}$ so far in the examples $r = 3, N = 5$

$$n_{total\ var} = n_{slow\ var} + n_{fast\ var}$$

$$n_{slow\ var}(p, r, N, t_f, \Delta t) = 2\, r \left(\frac{t_f}{p\, \Delta t} + 1\right)$$

$$n_{fast\ var}(r, N, t_f, \Delta t) = 2\,(N + 1 - r)\left(\frac{t_f}{\Delta t} + 1\right) + \frac{t_f}{\Delta t}$$

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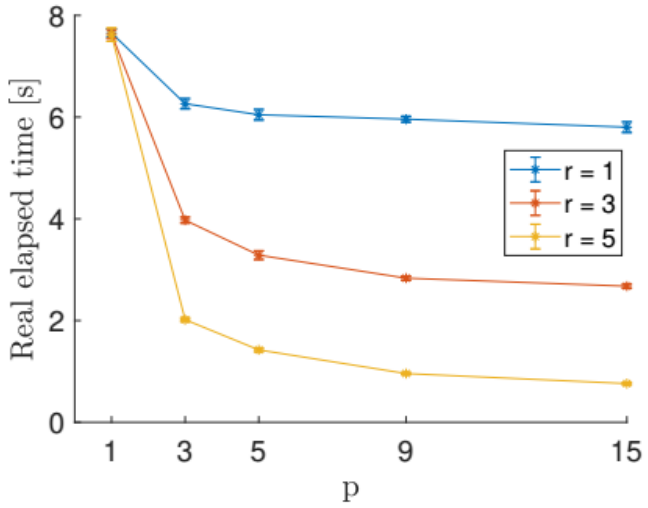


Figure 6: Mean computational time with standard deviation versus p for a constant micro time step of 10^{-3}, $t_f = 4.5s$ and $\theta_{t_f} = 20°$

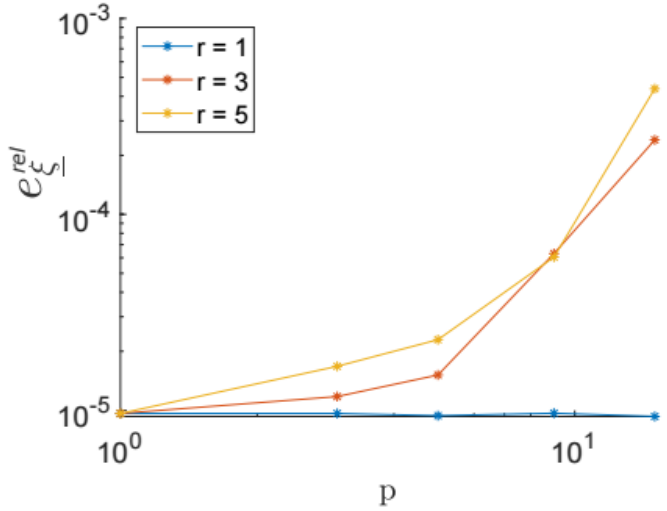


Figure 7: Relative error in $\underline{\xi}$ versus p for a constant micro time step of 10^{-3}, $t_f = 4.5s$ and $\theta_{t_f} = 20°$

Conclusions

- **Multirate Discrete Mechanics and Optimal Control (Multirate DMOC):**

  High fidelity simulation at a reduced computational cost

  - ✓ Structure-preserving -> *conservation of energy and/or momenta of the system*
  - ✓ Multirate discretization -> *computational efficiency*
  - ✓ Unified control methodology -> *potential improvement in optimality and constraint- handling capabilities*
  - ✓ State space formulation -> *further great computational saving for equivalent accuracy with structure preservation*
  - ✓ No need for decoupling of the equations of motion -> *simple application to a variety of linear and nonlinear models*
  - ✓ Straightforward selection of slow and fast subsystems -> *capability to tailor the solution to the time scales of the dynamics in question to obtain further reductions in computational cost*

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Flexibility to tailor the method to the time scales of the problem and obtain the required fidelity at a reduced computational cost!

Conclusions

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  High fidelity simulation at a reduced computational cost

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  Flexibility to tailor the method to the time scales of the problem and obtain the required fidelity at a reduced computational cost!

- **Future work –** methods for obtaining optimal p and r

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THANK YOU FOR YOUR ATTENTION!

Contact details:

Yana Lishkova, yana.lishkova@eng.ox.ac.uk

Address: St. Hugh's College, St Margaret's Rd, Oxford, OX2 6LE, United Kingdom

Original Paper:

[1] Y. Lishkova, S. Ober-Blöbaum, M. Cannon, and S. Leyendecker, "A multirate variational approach to simulation and optimal control for flexible spacecraft*", in AAS Advances in the Astronautical Sciences, Volume 175, Astrodynamics 2020*

Main references:

[2] J. E. Marsden and M. West, "Discrete mechanics and variational integrators," Acta Numerica, Vol. 10, No. 1, 2001, pp. 357–514.

[3] S. Leyendecker and S. Ober-Blöbaum, "A Variational Approach to Multirate Integration for Constrained Systems," *Multibody Dynamics: Computational Methods and Applications* (J.-C. Samin and P. Fisette, eds.), pp. 97–121, Dordrecht: Springer Netherlands, 2013

[4] T. Gail, S. Ober-Blöbaum, and S. Leyendecker, "Variational multirate integration in discrete mechanics and optimal control," *Proceedings of* ECCOMAS, 2017, pp. 1–4.

[5] O. Junge, J. E. Marsden, and S. Ober-Blöbaum, "Discrete mechanics and optimal control," *IFAC Proceedings Volumes*, Vol. 38, No. 1, 2005, pp. 538–543.

[6] S. Ober-Blöbaum, O. Junge, and J. E. Marsden, "Discrete mechanics and optimal control: an analysis," *ESAIM: Control, Optimisation and Calculus of Variations*, Vol. 17, No. 2, 2011, pp. 322–352.

[7] T. Gail, S. Leyendecker, and S. Ober-Blöbaum, "Computing time investigations of variational multirate integrators," *ECCOMAS Multibody Dynamics*, 2013.

[8] M. Azadi, M. Eghtesad, S. Fazelzadeh, and E. Azadi, "Dynamics and control of a smart flexible satellite moving in an orbit," *Multibody System Dynamics*, Vol. 35, No. 1, 2015, pp. 1–23.

[9] J. L. Junkins and Y. Kim, *Introduction to dynamics and control of flexible structures*. American Institute of Aeronautics and Astronautics, 1993.

The extension

The work so far has been published in the following paper:

> **Y. Lishkova, S. Ober-Blöbaum, M. Cannon, and S. Leyendecker, "**A multirate variational approach to simulation and optimal control for flexible spacecraft", *accepted for publication in Proceedings of AAS/AIAA Astrodynamics Specialist Conference, 2020*

Since this work was completed, we continued to explore other formulations of the MDMOC which are capable of even further great computational savings.

- with DMOC one can formulate the optimal control problem to avoid computation of the momentum/velocity explicitly while maintaining the conservational properties of the representation

- compared to standard techniques this further reduces the number of the number of constraints by approximately a half and the number of unknowns by more than a third with the exact amount determined by the formulation of the controls!

Comparison for simple OCP example

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Single rate DMOC

$$\min_{q_d,u_d,(h)} J_d(q_d,u_d) = \sum_{k=0}^{N-1} C_d(q_k,q_{k+1},u_k)$$

$$D_2L_d(q_{k-1},q_k) + D_1L_d(q_k,q_{k+1}) + f_{k-1}^+ + f_k^- = 0 \quad \text{for } k = 1,\dots,N-1$$

$$q_0 = q^0,\ D_2L(q^0,\dot{q}^0) + D_1L_d(q_0,q_1) + f_0^- = 0$$

Multirate DMOC

$$\min_{\underline{q}_d^s,\underline{q}_d^f,\underline{u}_d^{slow},\underline{u}_d^{fast}} J_d(\underline{q}_d^s,\underline{q}_d^f,\underline{u}_d^{slow},\underline{u}_d^{fast}) = \sum_{k=0}^{N-1} C_d^k(\underline{q}_k^s,\underline{q}_{k+1}^s,\underline{q}_k^f,\underline{u}_{k+1/2}^s,\underline{u}_k^f)$$

$$D_{\underline{q}_k^s}(L_d^k + L_d^{k-1}) + F_k^{s-} + F_{k-1}^{s,+} = 0 \qquad \text{for } k = 1,\dots,N-1$$

$$D_{\underline{q}_k^{f,0}}(L_d^k - L_d^{k-1}) + \underline{f}_k^{f,0,-} + \underline{f}_k^{f,p-1,+} = 0 \qquad \text{for } k = 1,\dots,N-1$$

$$D_{\underline{q}_k^{f,m}}L_d^k + \underline{f}_k^{f,m,-} + \underline{f}_k^{f,m-1,+} = 0 \qquad \text{for } k = 0,\dots,N-1,\ m = 1,\dots,p-1$$

$$\underline{q}_0^s = \underline{q}^s(0), \quad \underline{q}_0^{f,0} = \underline{q}^f(0),$$

$$\underline{p}_0^s = -D_{\underline{q}_0^s}L_d^0(\underline{q}_0^s,\underline{q}_1^s,\underline{q}_0^f) - F_0^{s,-} \qquad \underline{p}_0^{f,0} = -D_{\underline{q}_0^{f,0}}L_d^{0,0}(\underline{q}_0^{s,0},\underline{q}_1^s,\underline{q}_0^{f,0},\underline{q}_0^{f,1}) - \underline{f}_0^{f,0,-}$$

where $q = r(q^s,q^f)$ **and are controls known to have slow and fast dynamics respectively**

Comparison

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Standard direct methods

action integral

$$\int_0^T L(q(t), \dot{q}(t))\, \mathrm{d}t$$

virtual work

$$\int_0^T f(q(t), \dot{q}(t)) \cdot \delta q(t)\, \mathrm{d}t$$

Lagrange- d'Alembert principle

$$\delta \int_0^1 L(q(t), \dot{q}(t))\, dt + \int_0^1 f(t) \cdot \delta q(t)\, dt = 0$$

for all variations δq with $\delta q(0) = \delta q(1) = 0$

the *forced Euler-Lagrange equations*

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = f(q, \dot{q})$$

discretize the
forced Euler-Lagrange equations

Single rate DMOC

the *discrete Lagrangian*

$$L_d(q_k, q_{k+1}) \approx \int_{kh}^{(k+1)h} L(q(t), \dot{q}(t))$$

virtual work approximation
by the left and right discrete forces

$$f_k^- \cdot \delta q_k + f_k^+ \cdot \delta q_{k+1} \approx \int_{kh}^{(k+1)h} f(t) \cdot \delta q(t)\, dt$$

discrete Lagrange- d'Alembert principle

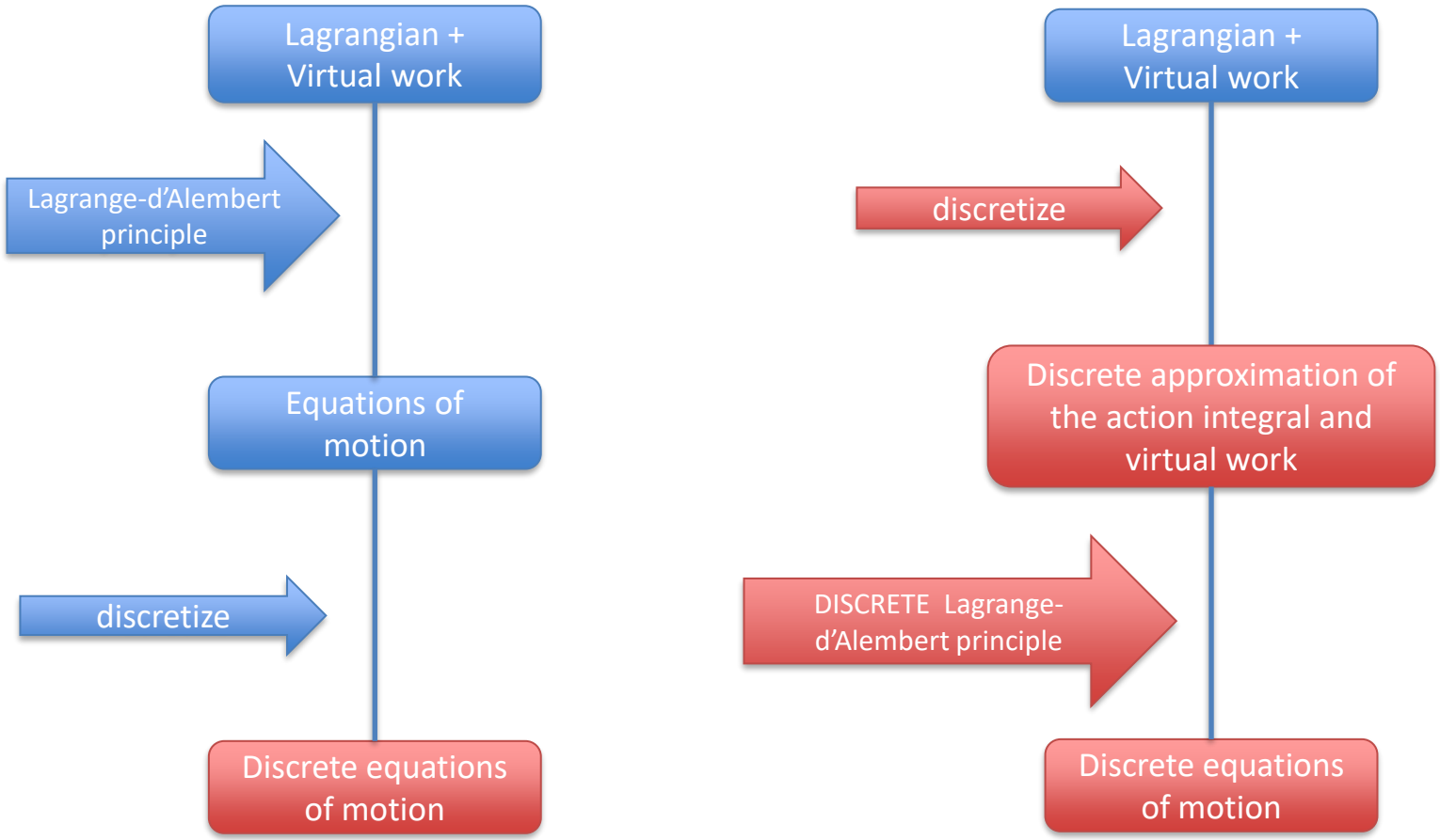
$$\delta \sum_{k=0}^{N-1} L_d(q_k, q_{k+1}) + \sum_{k=0}^{N-1} f_k^- \cdot \delta q_k + f_k^+ \cdot \delta q_{k+1} = 0$$

for all variations $\{\delta q_k\}_{k=0}^N$ with $\delta q_0 = \delta q_N = 0$

discrete forced Euler-Lagrange equations

$$D_2 L_d(q_{k-1}, q_k) + D_1 L_d(q_k, q_{k+1}) + f_{k-1}^+ + f_k^- = 0$$

Comparison


Standard direct methods
Lagrangian + Virtual work
Lagrange-d'Alembert principle
Equations of motion
discretize
Discrete equations of motion
Single rate and Multirate DMOC
Lagrangian + Virtual work
discretize
Discrete approximation of the action integral and virtual work
DISCRETE Lagrange-d'Alembert principle
Discrete equations of motion

Optimal control problem formulation

$$J(x,u) = \int_{t_0}^{t_f} C(\underline{x}(t), u(t))\, dt = \frac{1}{2}\int_{t_0}^{t_f} [\underline{x}(t)^T \mathbf{W} \underline{x}(t) + u(t)^2]\, dt$$

$$\text{subject to} \quad \begin{cases} \dot{\underline{x}}(t) = \mathbf{A}\underline{x}(t) + \mathbf{B}\tau(t) \\ \underline{q}(t_0) = \mathbf{E}^{-1}\,\underline{\xi}_{t_0}, \quad \underline{\xi}_{t_0} = [0, ..., 0]^T \\ \underline{q}(t_f) = \mathbf{E}^{-1}\,\underline{\xi}_{t_f}, \quad \underline{\xi}_{t_f} = [\theta_{t_f}, 0, ..., 0]^T \\ \dot{\underline{q}}(t_0) = \dot{\underline{q}}(t_f) = [0, ..., 0]^T \end{cases}$$

$$\text{where} \quad \underline{x}(t) = \begin{bmatrix} \underline{q}(t) \\ \dot{\underline{q}}(t) \end{bmatrix}, \quad u(t) = \tau(t), \quad \mathbf{A} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{\Lambda} & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \mathbf{E}^T\mathbf{D} \end{bmatrix}, \quad \mathbf{W} = \mathbf{I}$$

In this example the Legendre transform takes the following simple form:

$$\underline{p} = \frac{\partial \mathcal{L}}{\partial \dot{\underline{\xi}}} = \mathbf{M}\,\dot{\underline{\xi}}$$

transformation to modal coordinates [7]

$$\underline{\mathfrak{p}} = \frac{\partial \mathcal{L}}{\partial \dot{\underline{q}}} = \dot{\underline{q}}$$

Advantages

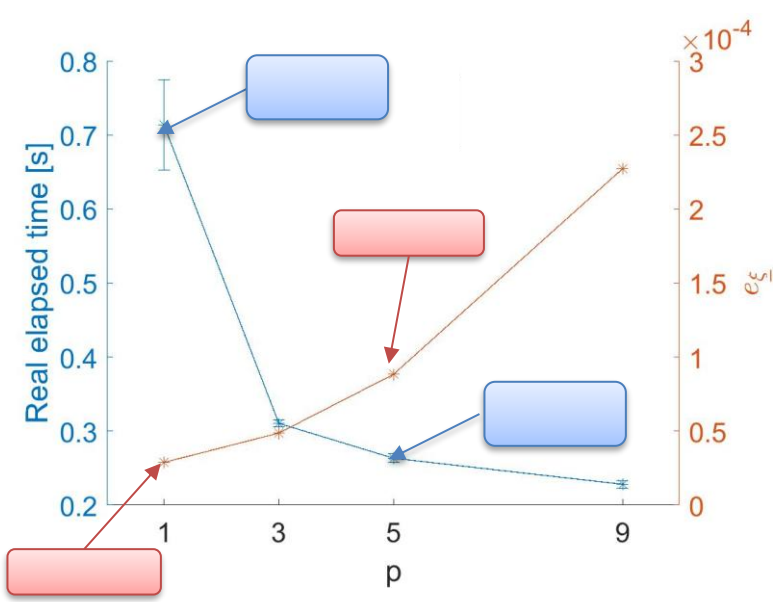
$$W_d(\underline{q}_d^s, \underline{q}_d^f, \underline{u}_d) = \sum_{k=0}^{N-1} (W_{d,k}^s(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f, \underline{u}_k) + W_{d,k}^f(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f, \underline{u}_k))$$

$$\underline{f}\,(\underline{q}^s, \underline{q}^f) \cdot (\delta\underline{q}^s, \delta\underline{q}^f) = \underline{f}^s \cdot \delta\underline{q}^s + \underline{f}^f \cdot \delta\underline{q}^f$$

$$L_d^k = L_d^k(\underline{q}_k^s, \underline{q}_{k+1}^s, \underline{q}_k^f) = \sum_{m=0}^{p-1} L_d^{k,m}(\underline{q}_k^{s,m}, \underline{q}_k^{s,m+1}, \underline{q}_k^{f,m}, \underline{q}_k^{f,m+1}) \approx \int_{t_k}^{t_{k+1}} \mathcal{L}(\underline{q}^s, \dot{\underline{q}}^s, \underline{q}^f, \dot{\underline{q}}^f)dt$$

Main result- the trade-off:

Formulation with momentum computation



Formulation without momentum computation

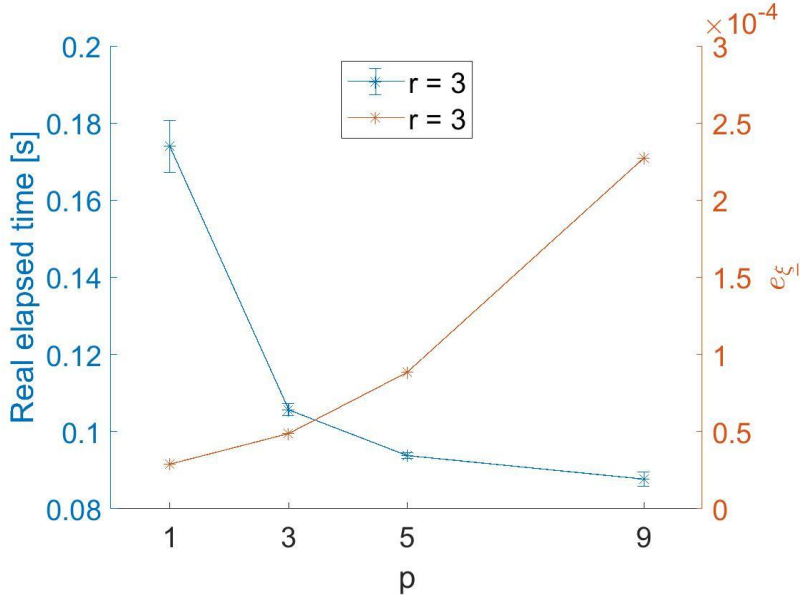


Figure 8: Mean computational time with standard deviation and relative error in ξ versus p for constant micro time step $\Delta t = 10^{-4}$, $t_f = 0.45s$ and no cost penalty for the momentum $\left(\mathbf{W} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & 0 \end{bmatrix} \right)$